Electronic and Telecommunication Engineering

Section 3

Computed Tomography (CT)

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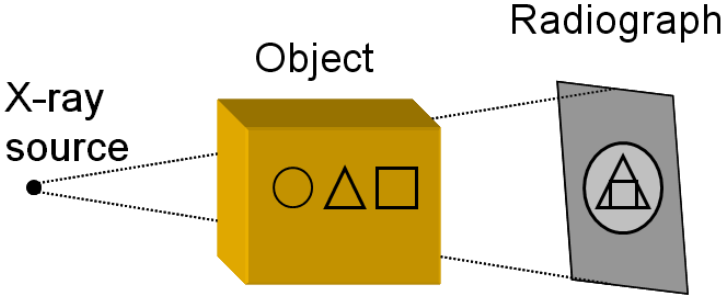
Acknowledgement: Dr. Nuwan Dayananda for the content

BM3122- Medical Imaging

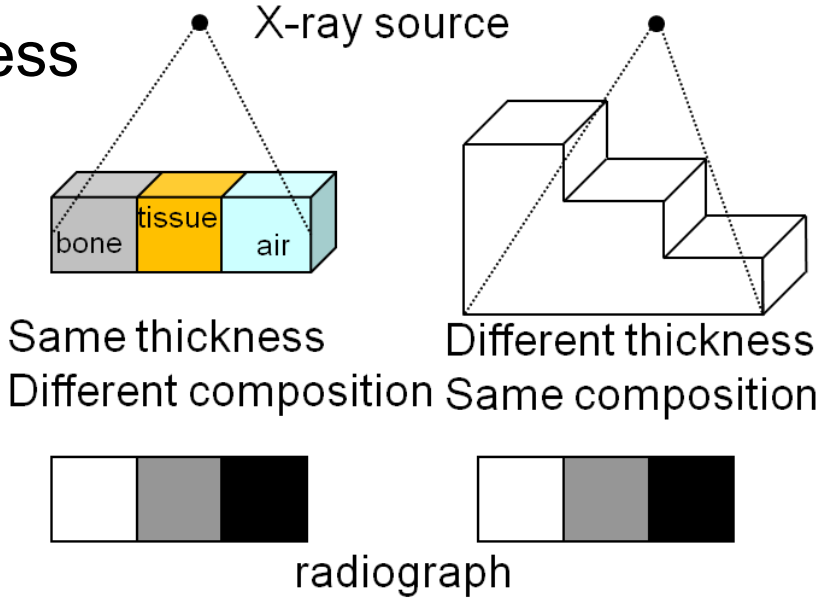
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Inadequacies of Projection Radiography

- Poor low contrast sensitivity (eg: muscle & fat)
- Superposition of 3D structures onto 2D images



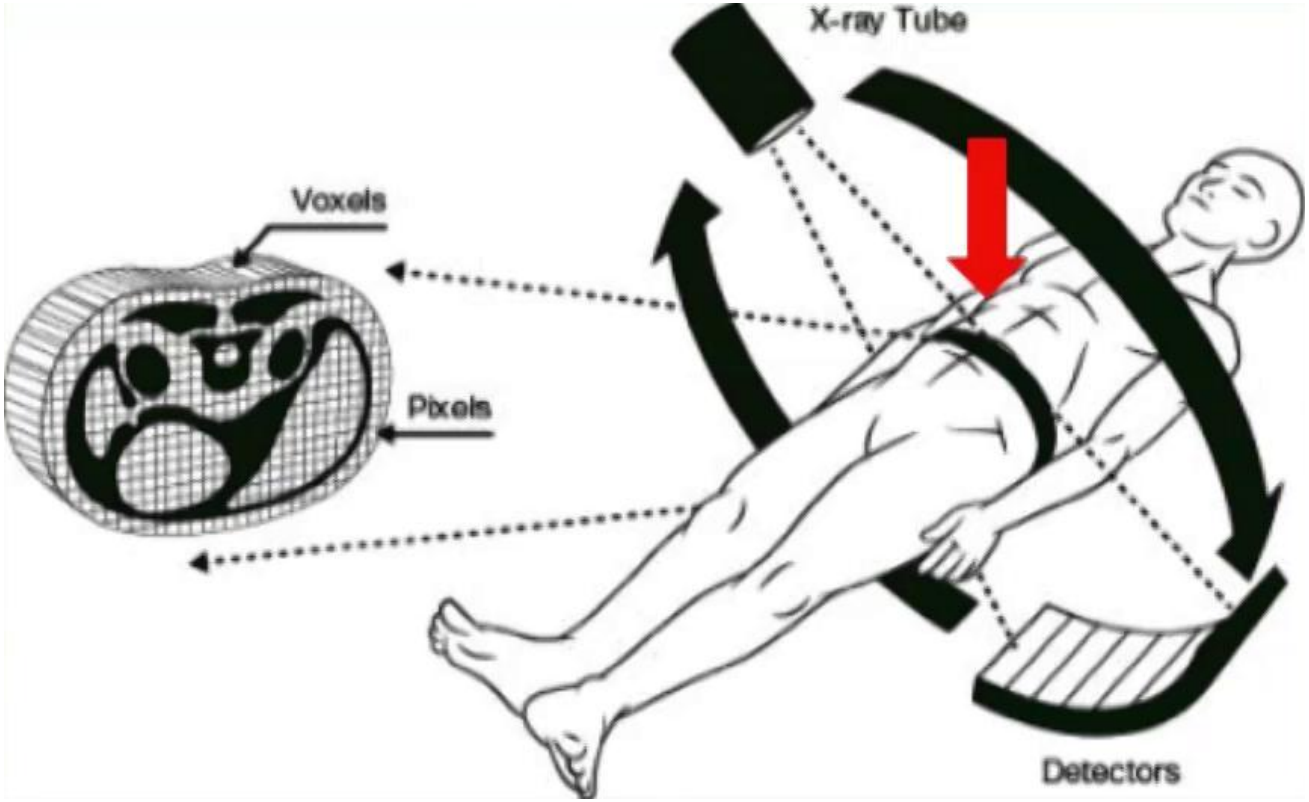
- Ambiguities with respect to thickness and tissue composition



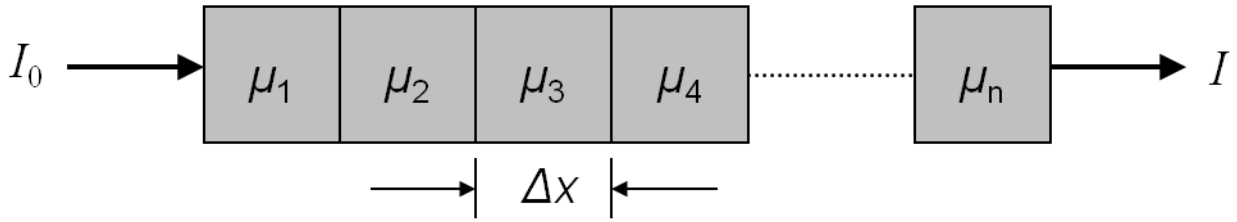
Inadequacies of Projection Radiography

Solution

Cross-sectional imaging with (X-ray) computed tomography (CT)



Basic Principle



- Absorption of incident X-ray (I_0) through the medium with linear attenuation coefficients, μ_i , i = 1,2,3,…,n, produces an X-ray intensity, I, at the detector.

Therefore, $I = I_0 e^{-\sum \mu_i \Delta x}$

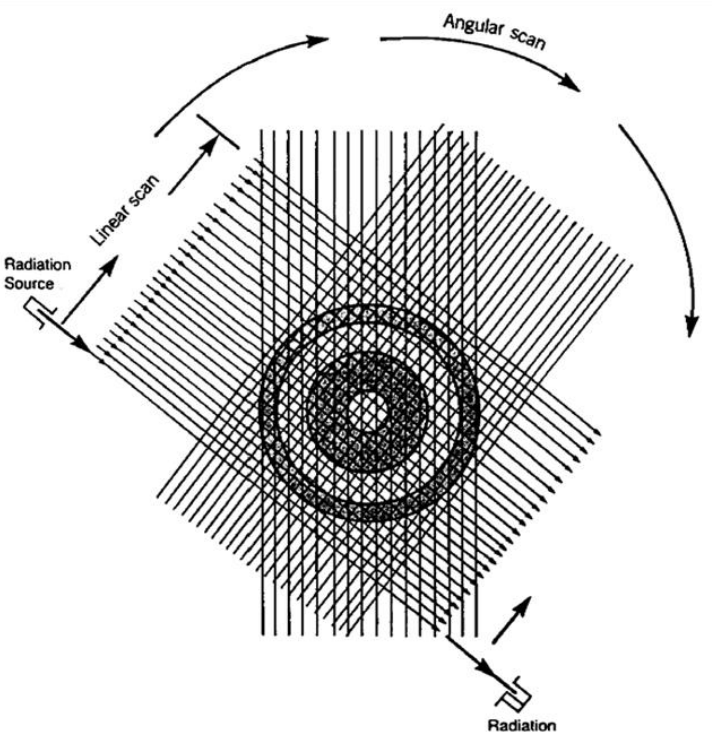
In the limit of small Δx, $I = I_0 e^{-\int_L \mu(x,y)dx}$

$\ln \frac{I_0}{I} = T = \int_L \mu(x,y)dx$, where $L = n\Delta x$

- Therefore, the distribution of $\mu(x,y)$ could be obtained, if we collect a sufficient number of values of T.

Basic Principle cont.

- Theoretical solution
  - A radiation source scans linearly along an object at a given view
  - Then, the source and detector pair rotate a small angle and repeat the linear scan
  - Continued until the completion of **180^0** rotation collects required no. of T values



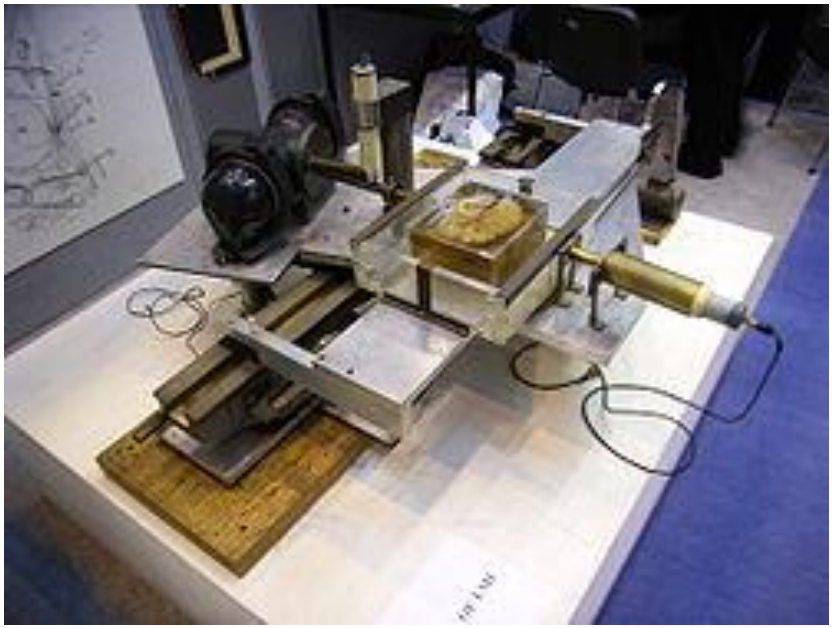
- Takes a long time

  → Not feasible

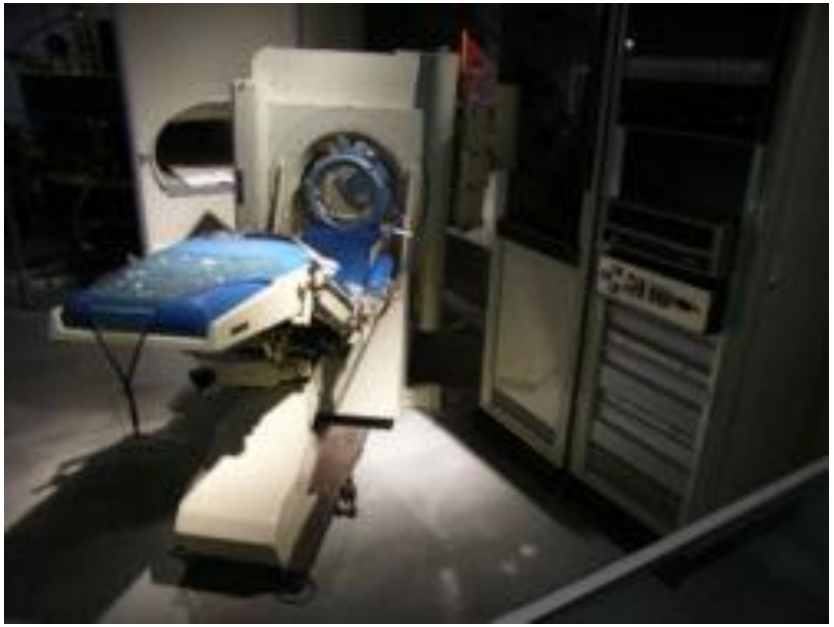
History

- Theoretical basis of reconstruction from projections developed by J. Radon (1917)
- Described independently by A. Cormack for radiation treatment planning (1963) and by G. Hounsfield (1967)
- Developed into clinical system by Hounsfield at EMI (Electro-Musical Instrument) (1972)
- Hounsfield and Cormack shared the Nobel Prize in Medicine (1979)

History cont.



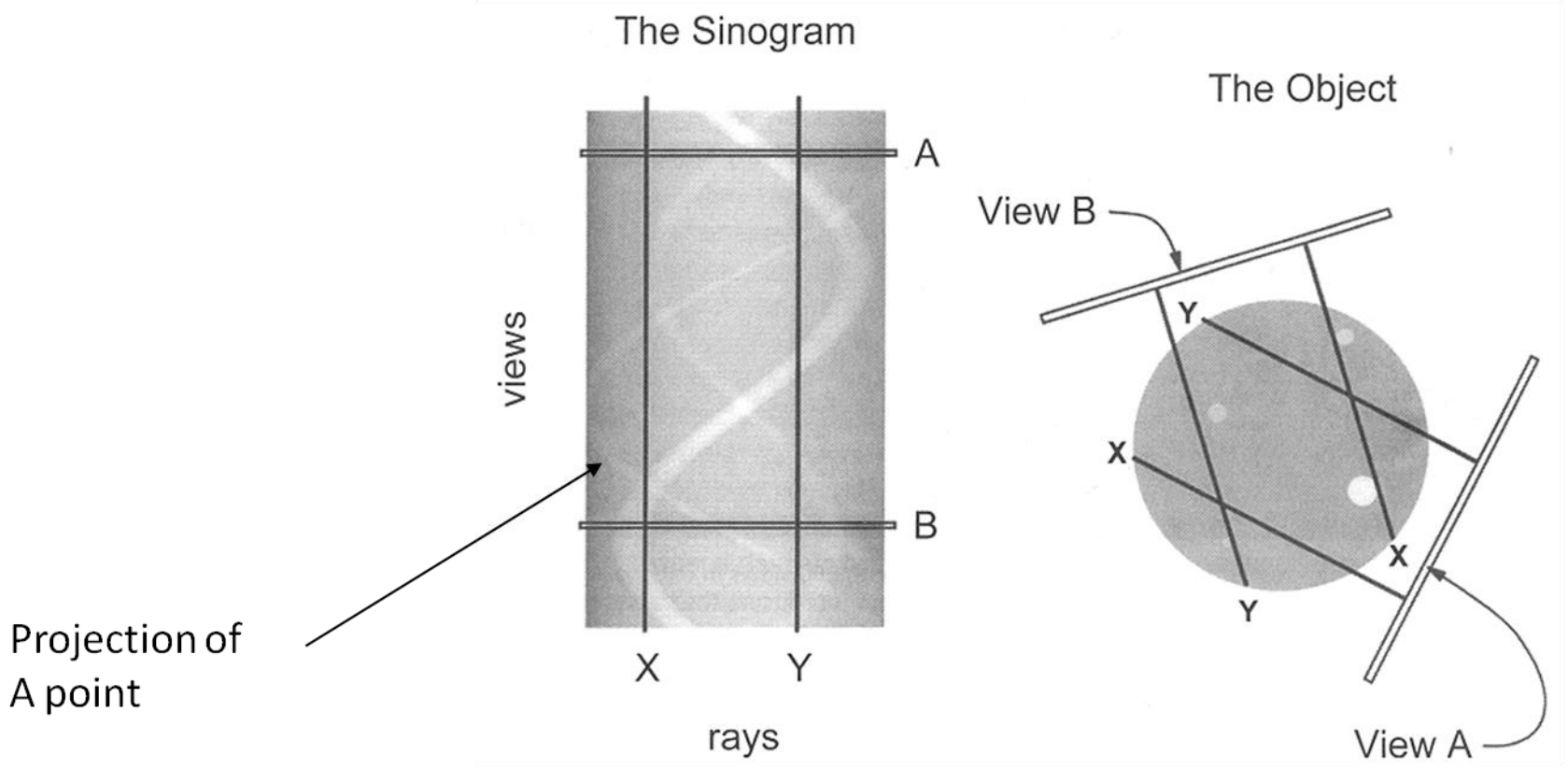
Hounsfield's prototype of CT scanner



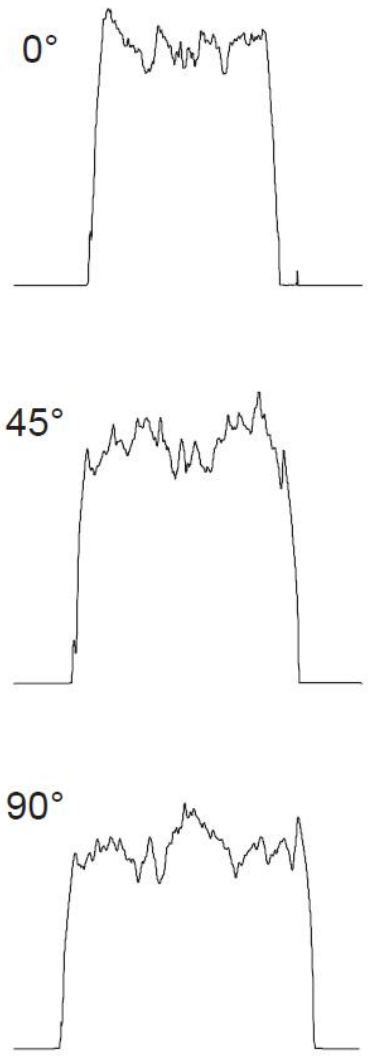
A historic EMI-Scanner

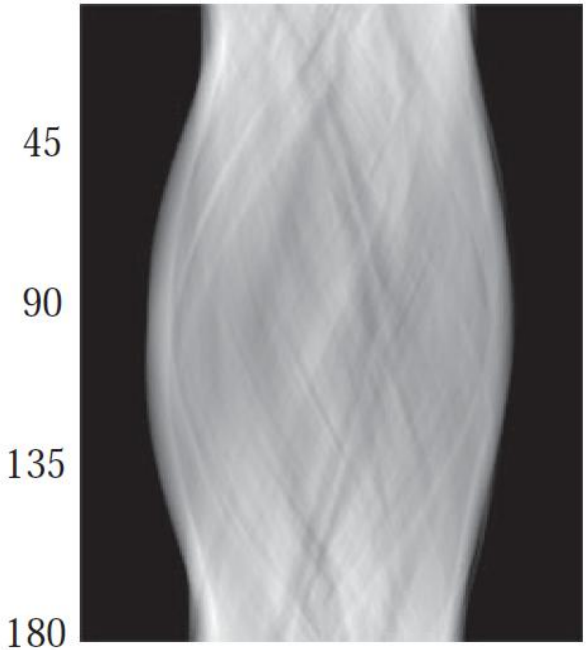
Sinograms

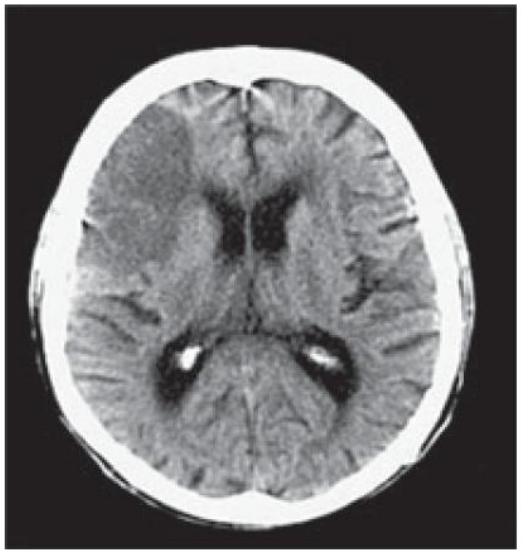
- Raw data collected before reconstruction



Sinograms cont.







Reconstruction Techniques

Image Reconstruction

- Reconstruction of a two-dimensional image from a series of one-dimensional projections
  - Required for CT imaging and also for PET and SPECT
  - A large number of one-dimensional projections, $p_1, p_2, \ldots, p_n$, are acquired at different angles, φ

Image Reconstruction cont.

- The object being imaged is represented as f (x, y)
  - Where x and y represent the image coordinates
  - The projection plots the intensity of the projection as a function of r

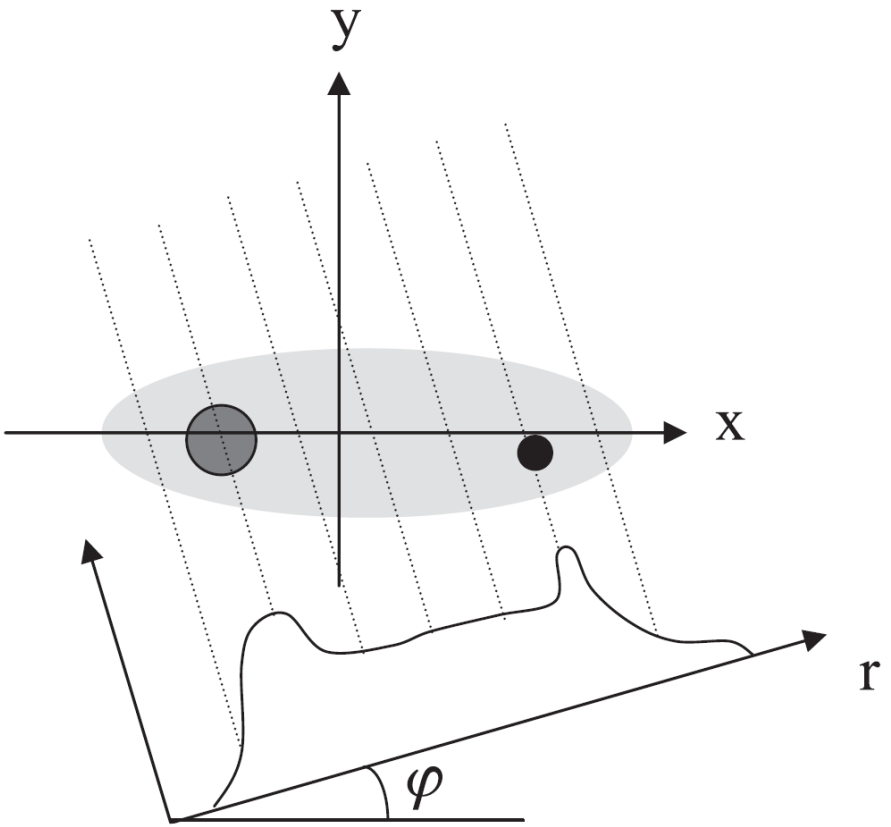


Image Reconstruction cont.

- Spatially-dependent values of $f(x, y)$ depends upon the imaging modality
  - Attenuation coefficients in X-ray CT
  - Distribution of radio-pharmaceutical in SPECT or PET

- The measured projection at every angle φ is denoted by $p(r, \varphi)$
  - In general, φ varies between 0° to 180°

Reconstruction Algorithms

Analytic reconstruction

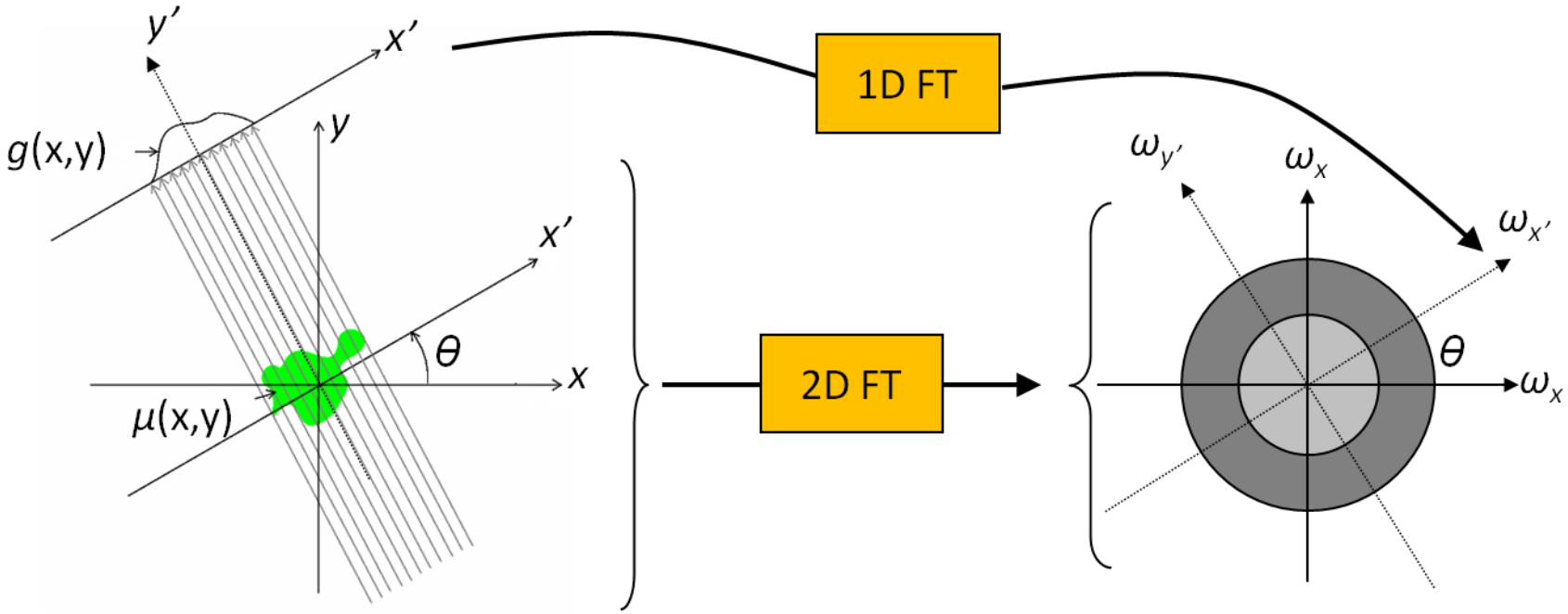
- Provides a direct mathematical solution for the image
- Fast
- Does not model the variations in detection process

Iterative reconstruction

- Allows more accurate modeling of the data acquisition
- Longer computation times
- Considered more suitable when full angular coverage is not available

Fourier Transform Technique

- Based on **central section theorem**
  - The 1D Fourier Transform (FT) of a projection at an angle θ equals to a line through the 2D FT of the object, at the same angle, θ



Fourier Transform Technique cont.

- Sufficient projection views are collected to fill the Fourier domain
  - *μ(x, y)* is obtained from **inverse FT**

- In practice, the direct FT technique is rarely used
  - Conceptually simple, but computationally complex
  - Polar sampling in Fourier domain requires interpolation before inverse FT
  - Entire projection data set is required prior to 2D inverse FT

Back Projection

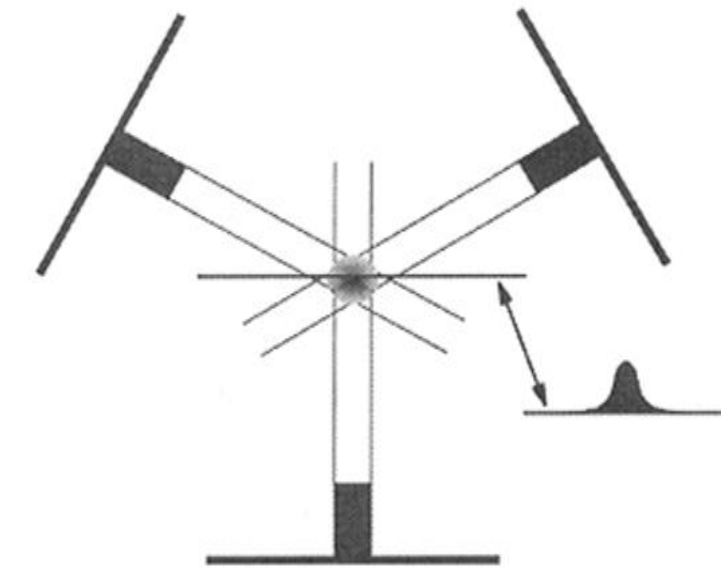
- Most commonly used form of reconstruction

- Assumptions:
  - **Linearity**
  - **Superposition**

    → Image can be reconstructed as the sum of many points

Back Projection cont.

- Projection data is “smeared" along path through the reconstruction plane
- Artifact results from “**tails**" of projection data



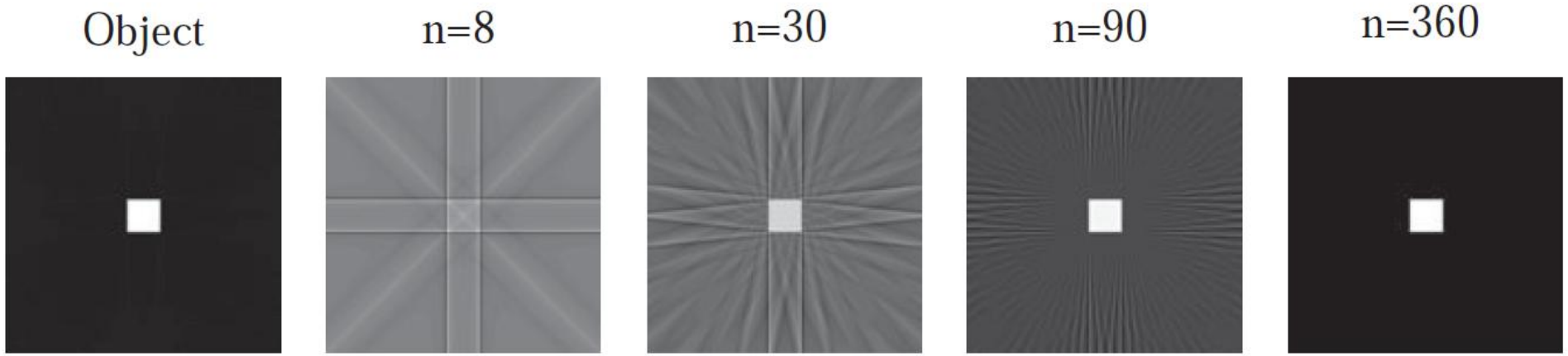
simple backprojection

Back Projection cont.

- The reconstructed image, $\hat{f}(x, y)$ of the object, $f\ (x, y)$ after back projection is given by:

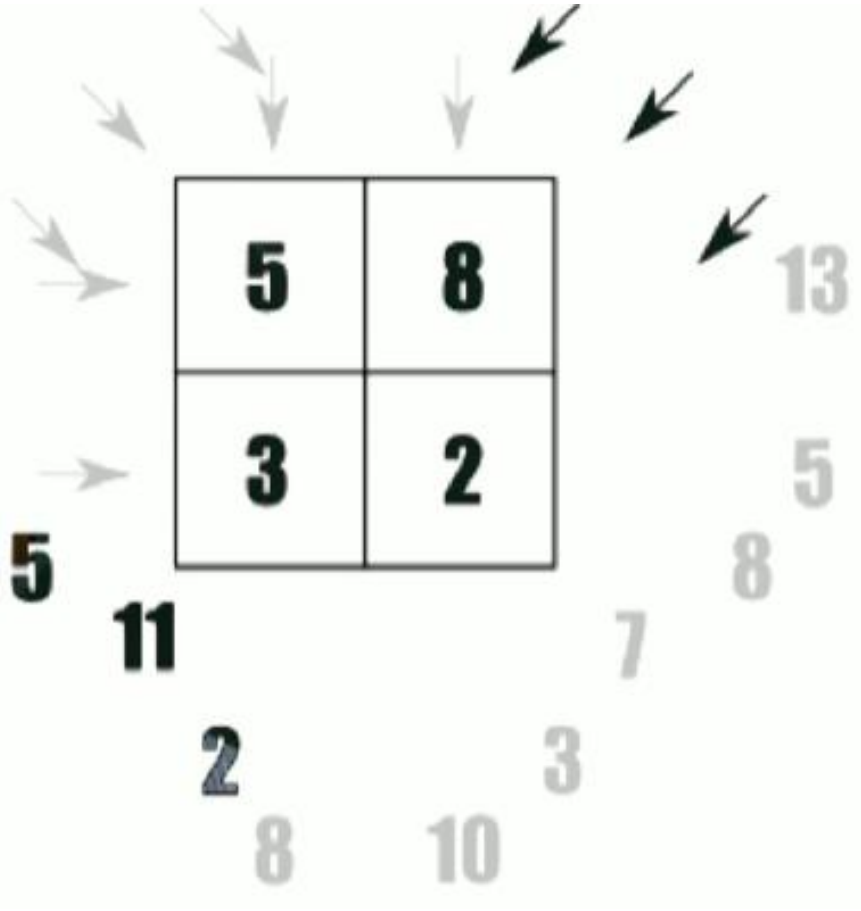
$$\hat{f}(x, y) = \sum_{j=1}^{n} p(r, \varphi_j)$$

Where n is the number of projections



Back Projection cont.

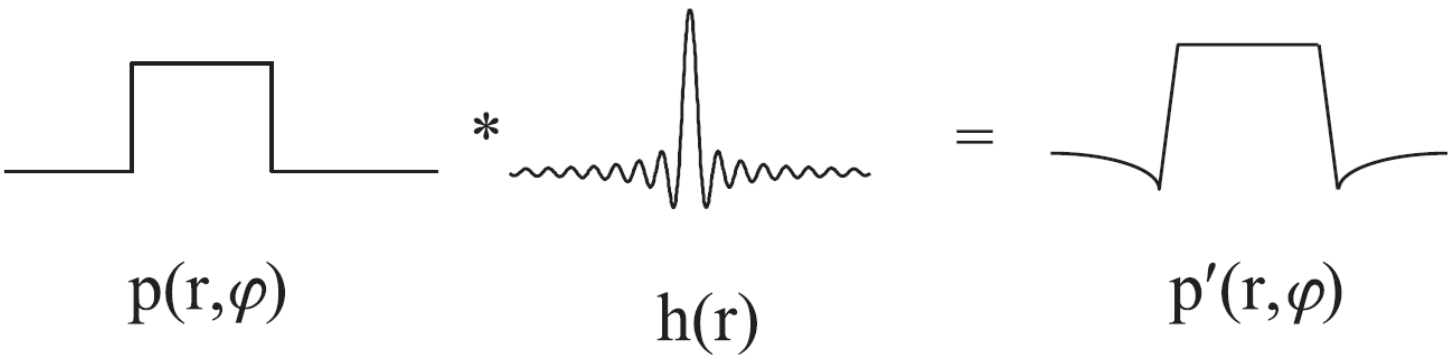
Numerical example:



Filtered Back Projection

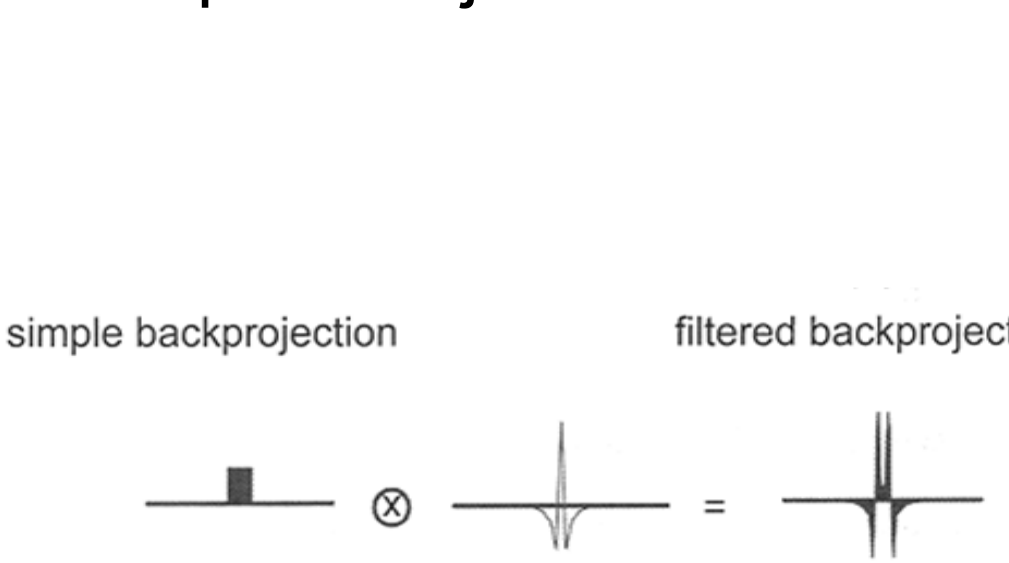
- Each projection is modified prior to back projection in order to account for the “tails"
- Correction can be performed by convolution with kernel of the appropriate shape
- Negative lobes are introduced around the edges of the projection
  - Cancel out the unwanted positive ‘blurring’ signal around the edge of the object
- The filtered projection in spatial domain;

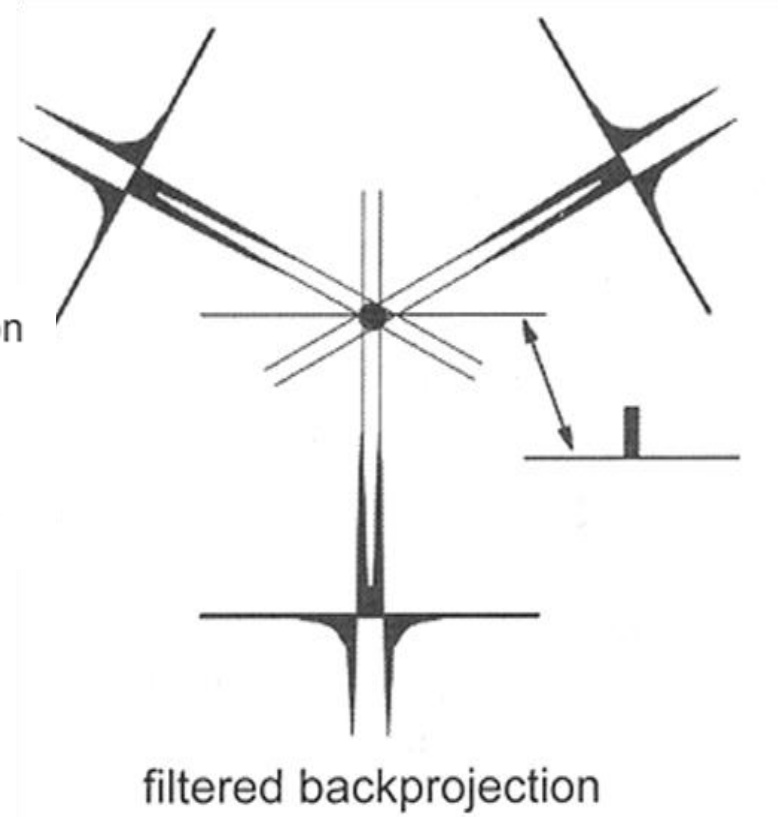
$$\acute{p}\left(r,\varphi_j\right) = p\left(r,\varphi_j\right) * h\left(r\right)$$



Filtered Back Projection cont.

- Kernel shape can be understood from consideration of point object





Filtered Back Projection cont.

- The filtered backprojection is normally carried out in the spatial frequency-domain using fast Fourier transform methods
  - Mathematical process of convolution is computationally intensive and slow
  - The filtered projections:

$$\acute{P}(k,\varphi) = P(k,\varphi\ )\ H(k)$$

Where $P(k,\varphi) = F\{\ p(r,\varphi)\}$ and $H(k) = F\{\ h(r)\}$

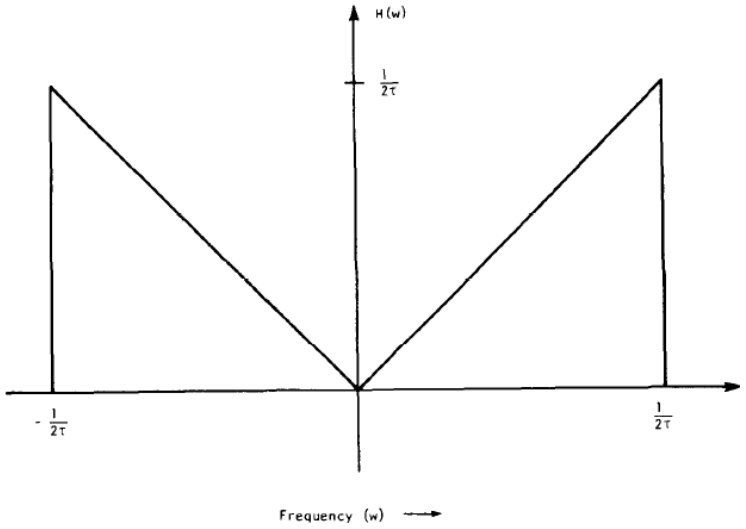
- The final filtered back projected image is given by:

$$\hat{f}(x,y) = \sum_{j=1}^{n} F^{-1}\{\ \grave{P}\left(k,\varphi_j\right)\}$$

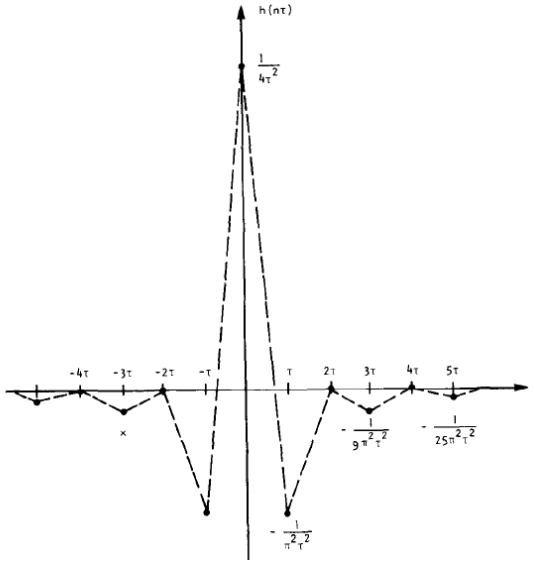
Filtered Back Projection cont.

$$H(w) = \begin{cases} |w|, & |w| < W \\ 0 & otherwise \end{cases}$$

w: spatial frequency



$$h(n\tau) = \begin{cases} 1/4\tau^2, n = 0 \\ 0, \quad n\ even \\ -\dfrac{1}{n^2\pi^2\tau^2}, n\ odd \end{cases}$$



- Apodizing window functions are typically applied to smooth the image
  - Hamming window
  - Hanning window
  - Cosine filter

Filtered Back Projection cont.

Example: filtered back projection using the Shepp-Logan phantom

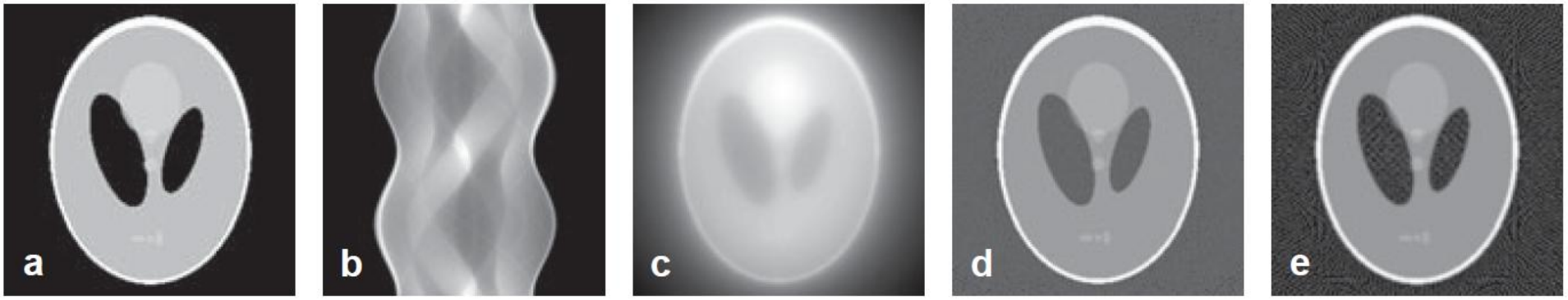
a: The Shepp-Logan phantom

b: The sinogram produced using 180 projections

c: Image reconstructed using simple backprojection with no filtering

d: The image reconstructed using filtered backprojection

e: Reconstruction with a band-pass filter



ART: Algebraic Reconstruction Technique

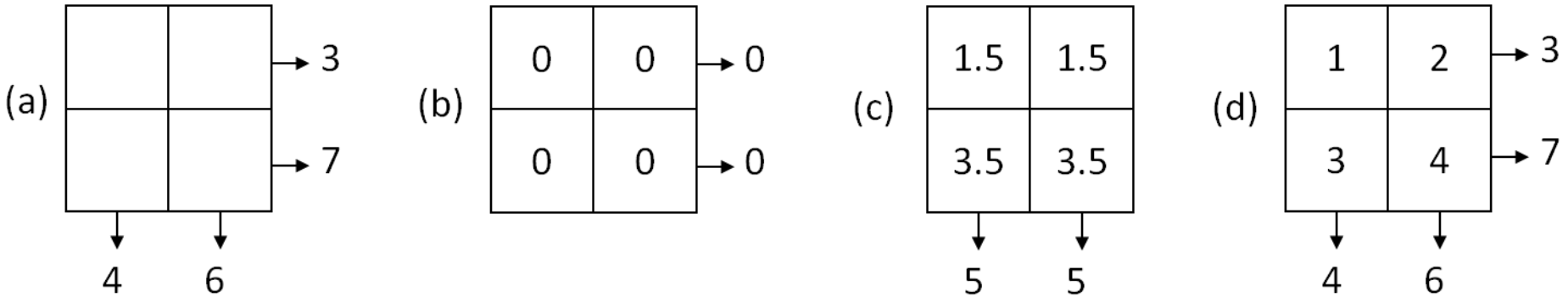
- Iterative approach
  - Used in the first CT scanner (EMI scanner)

- Basic principle:
  - If the estimated image is correct, then ray sums should equal measured projection values
  - Begin by assuming uniform values for $\mu_{i,j}$
  - Determine the difference between ray sums and measured projections
  - Distribute error uniformly along all pixels in that ray
  - Iterate: Apply to all rays at each angle until converges

ART: Algebraic Reconstruction Technique cont.

Example:

- Measure ray sums at all angles (a)
- Assume a uniform image with value zero (b)
- Consider angle 1 (horizontal summation), (a), divide the error evenly between all elements along the ray, (c)
- Consider angle 2 (vertical summation), (c), and divide the error evenly between all elements along the ray (d)
- Repeat the above step until converge (often leaving an acceptable error between observed and calculated values)



ART: Algebraic Reconstruction Technique cont.

- Advantages
  - Simple to code
  - Complete data set is not required to start

    → Can “estimate" image from very small number of views

- Disadvantages
  - Computational complexity (very slow)
  - Poor precision with noisy data

    → Resulting in oscillation, not convergence

MLEM Algorithm

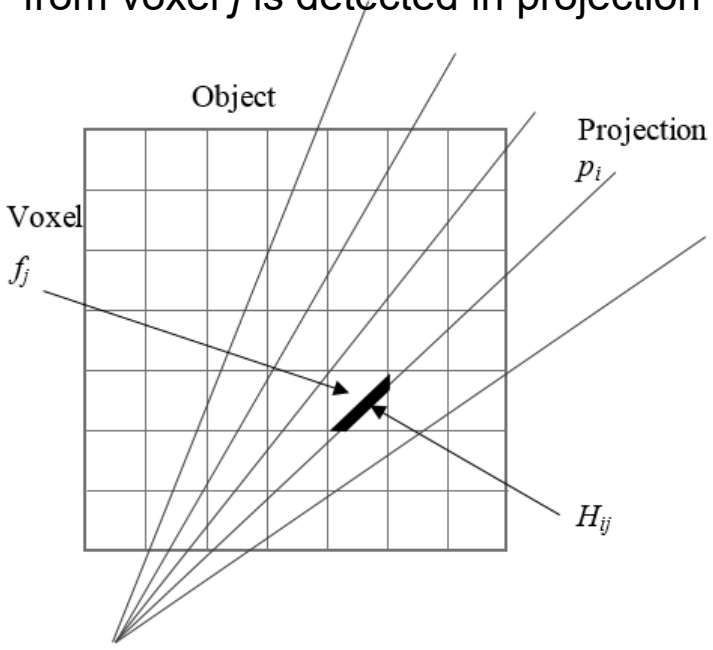
- Maximum-likelihood expectation-maximization (MLEM) algorithm

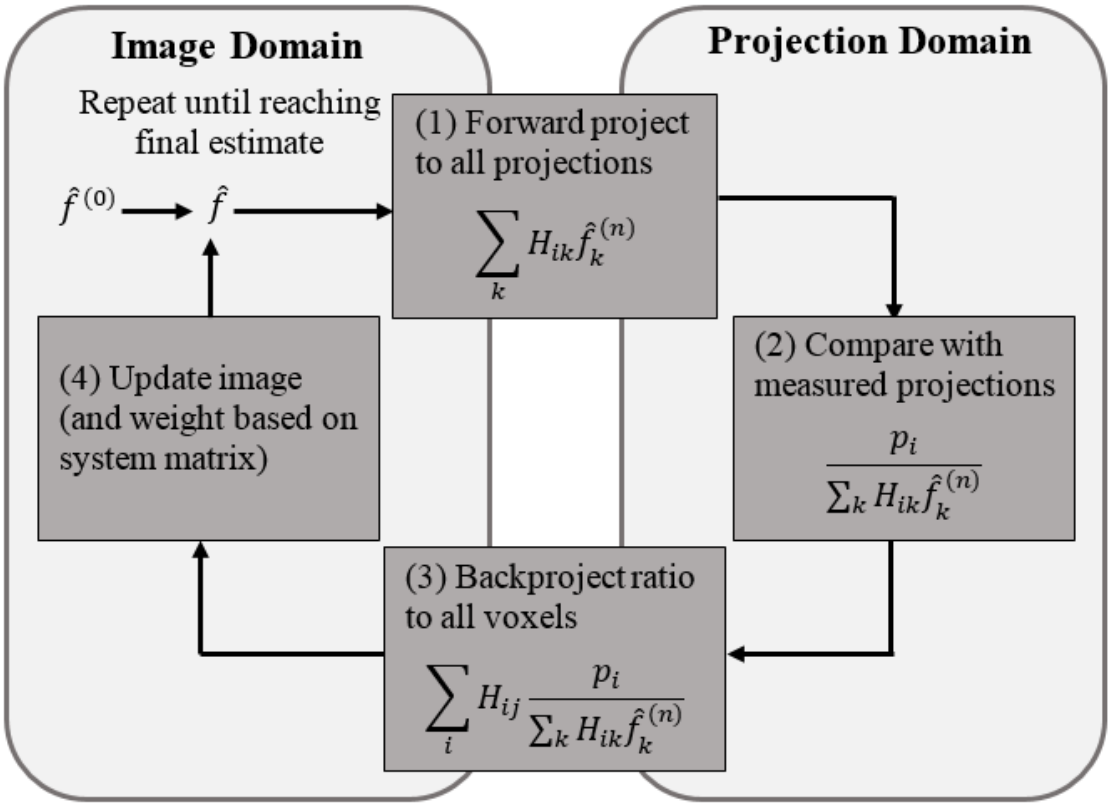
$$\hat{f}_j^{(n+1)} = \frac{\hat{f}_j^{(n)}}{\sum_{i`} H_{i`j}} \sum_i H_{ij} \frac{p_i}{\sum_k H_{ik} \hat{f}_k^{(n)}}$$

f_j : Value in voxel j

p_i : Value of the i[th] projection

H_{ij} : Probability that the contribution from voxel *j* is detected in projection *i*





CT Numbers (Hounsfield Units)

CT Numbers (Hounsfield Units)

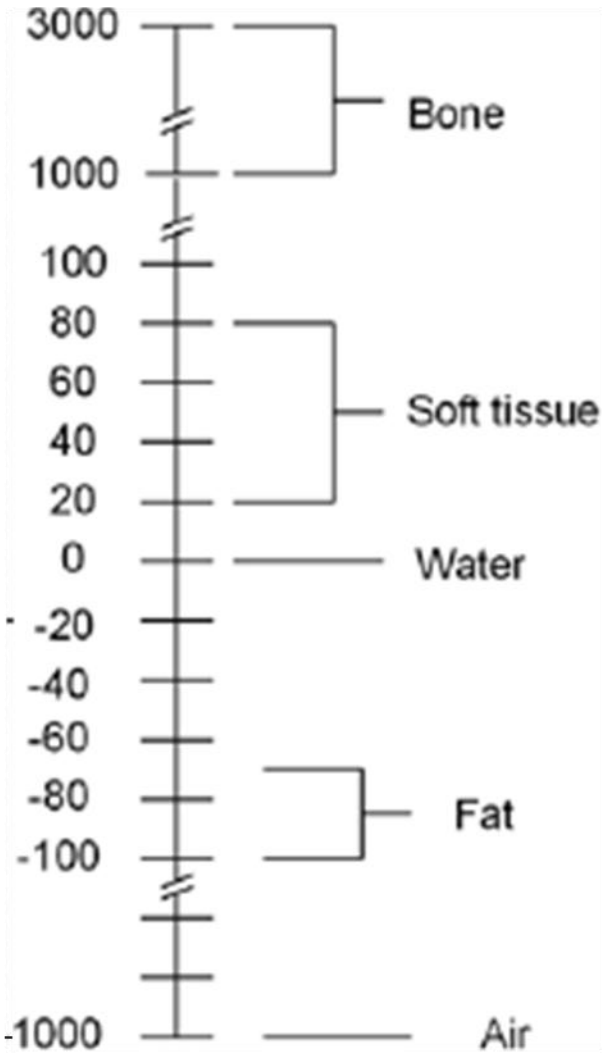
- CT numbers or Hounsfield units (HU) are related to linear attenuation coefficient of the corresponding material

$$HU(x,y) = 1000 \times \left[\frac{\mu(x,y) - \mu_{water}}{\mu_{water}}\right]$$

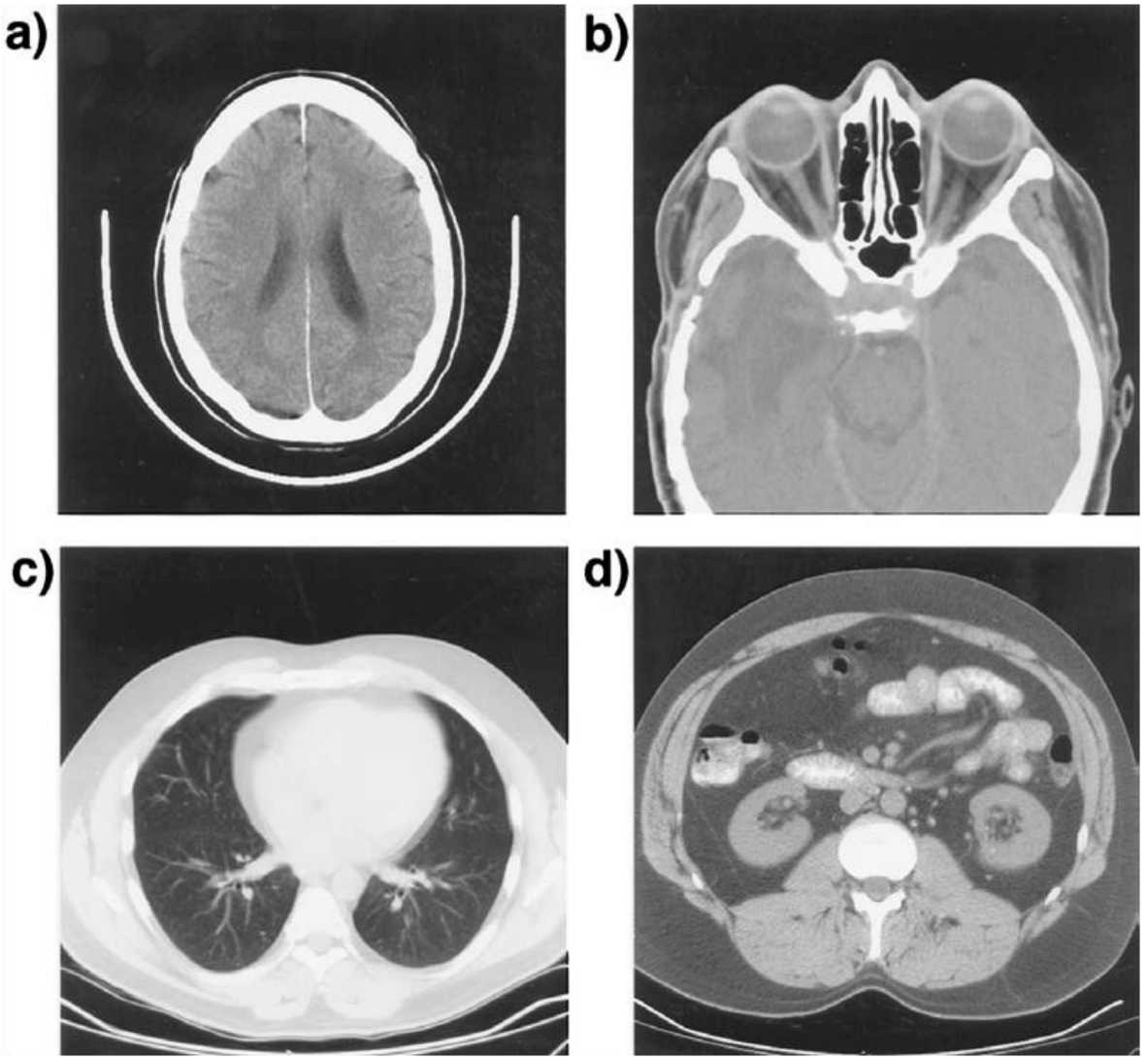
- CT numbers range from about -1,000 to +3,000
  - Air: -1,000
  - Soft tissues: range from -300 to 100
  - Water: 0
  - Bone and areas with contrast agents: up to +3,000

CT Numbers (Hounsfield Units) cont.

- CT numbers are quantitative
- CT scanners measure bone density with good accuracy
  - Can be used to assess fracture risk
- CT is also quantitative in terms of linear dimensions
  - Can be used to accurately assess tumor volume or lesion diameter



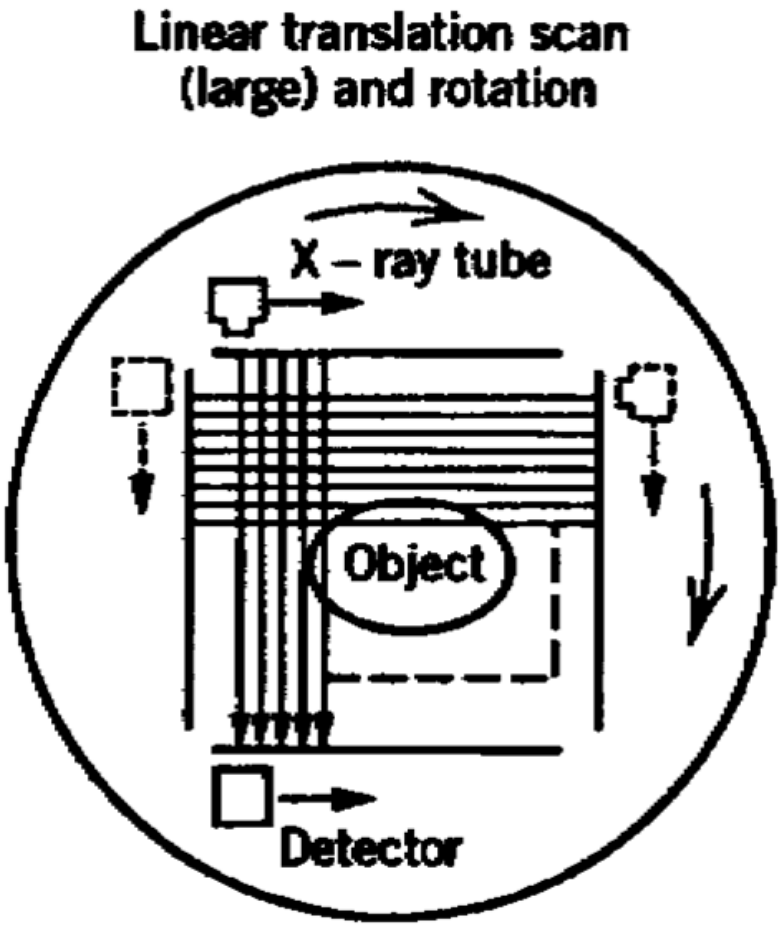
Typical CT Images



CT Scanner Designs

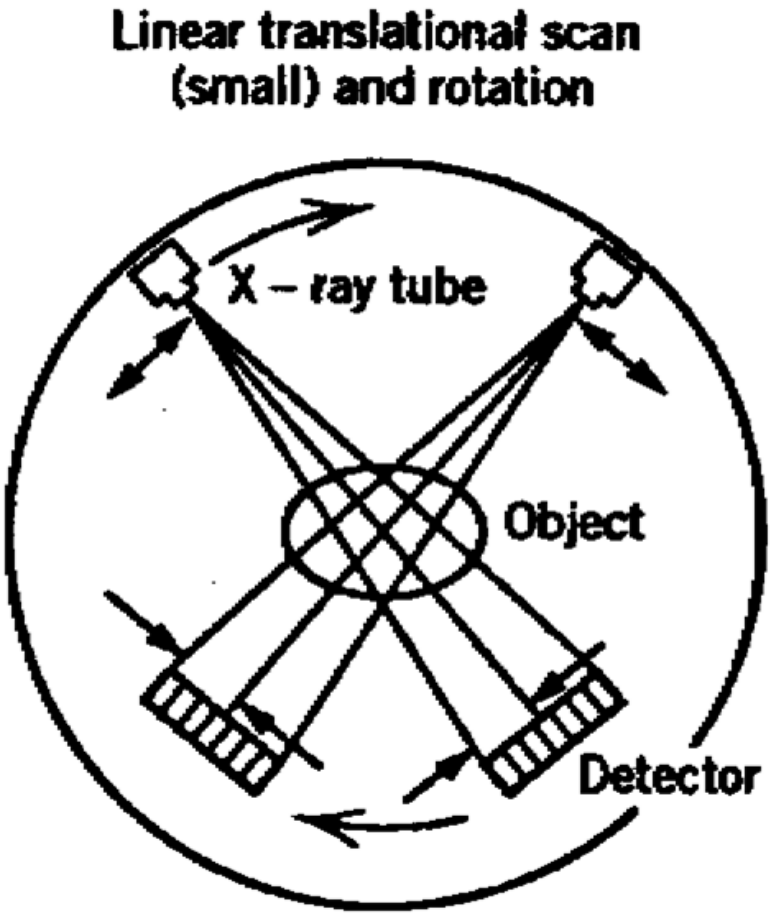
First Generation CT

- **Translate-Rotate**
- Mechanically simple
- Self-calibrating detectors
- Extremely slow (≈5min/scan)
- Inefficient: Wasting a lot of X-ray produced
- Never used in modern scanners



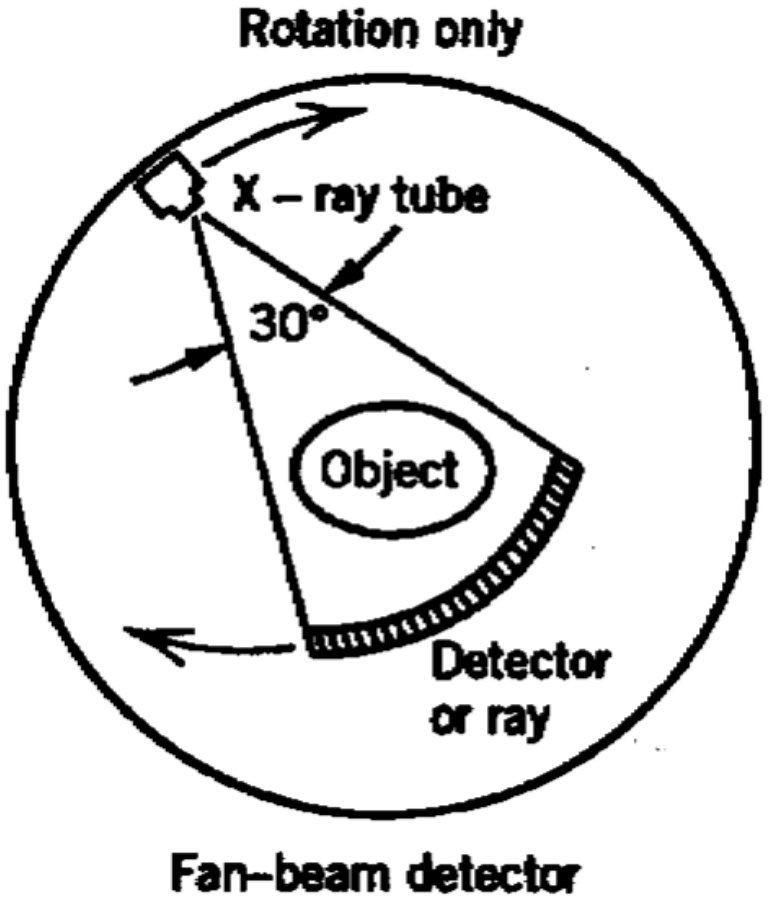
Second Generation CT

- **Translate (small)-Rotate**
- Similar to first generation, but with multiple detectors
- Faster scans
- Retains concept of self calibrating detectors
- Seldom used in modern scanners



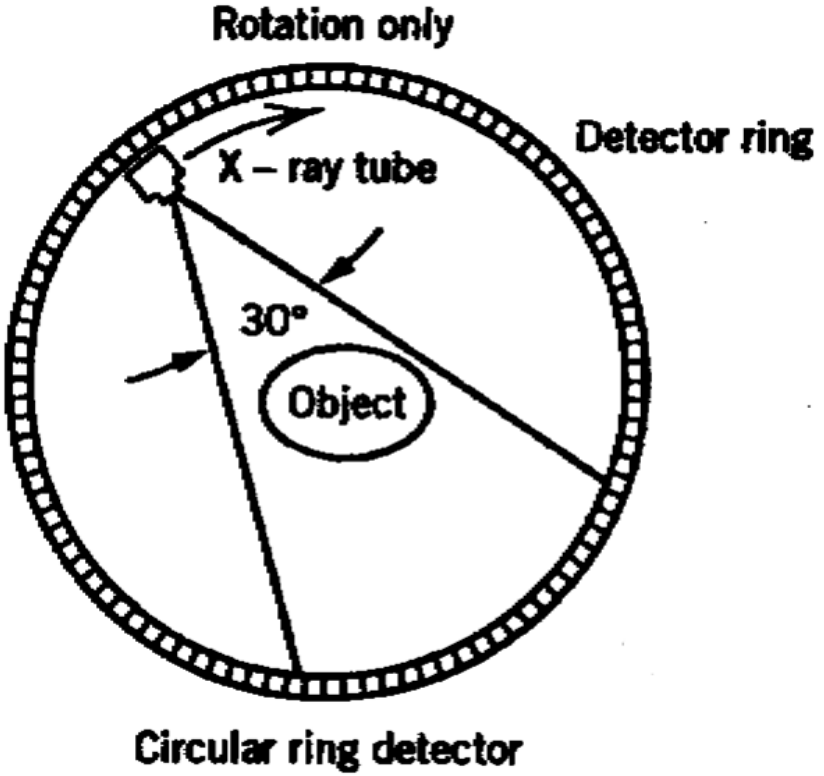
Third Generation CT

- **Rotate-Rotate**
- Multiple detectors cover entire “fan" beam through patient
- Both detectors and X-ray source rotate in synchrony
- Much faster (≈1sec/scan)
- Detector drift can be a problem (no calibration during the scan)
- Good detector collimation (grid) possible



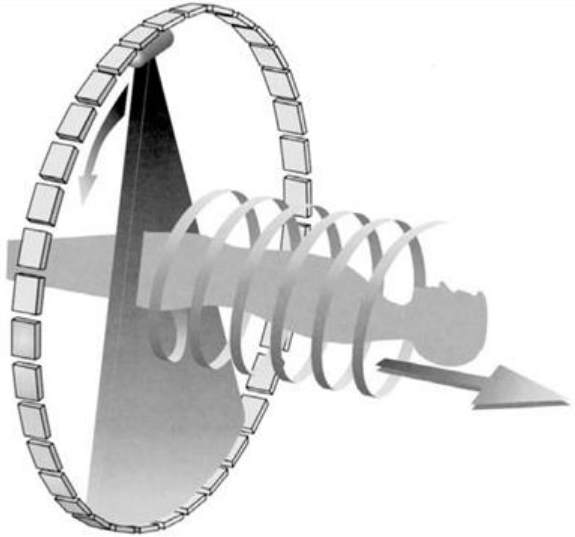
Fourth Generation CT

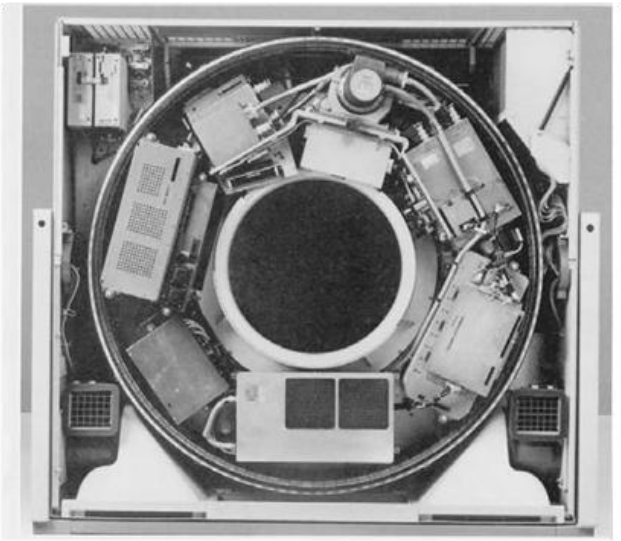
- **Rotate-Stationary**
- Only X-ray source rotates
- Detectors remain fixed
- Detector drift is not a problem
- Collimation with fixed collimators are not possible



Spiral CT (Helical CT)

- Applicable either to third or forth generation
- Rotating components are designed for continuous rotation
  - No cables to wind up
  - Slip rings
- Rotations can be as fast as two per second
- Patient couch is moved through scanner: To build up a 3D image





Spiral CT (Helical CT) cont.

- Special issues:
  - A ‘contactless’ method of both delivering power and receiving data had to be designed
  - Anode heating is much greater than for single-slice imaging
  - Beam pattern through the patient is now helical rather than consisting of parallel projections

- In a helical scan, the pitch (*p*) is defined as:

$$p = \frac{d}{s}$$

Where d = table feed per rotation of the X-ray tube

s = the collimated slice thickness

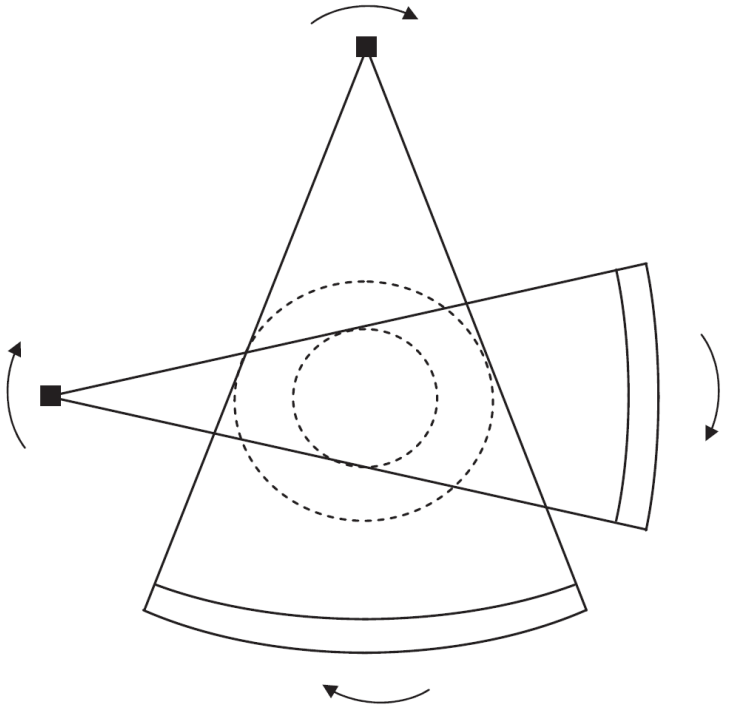
- The value of *p* typically used in clinical scans lies between 1 and 2
  - Equal to the reduction in tissue radiation dose compared to the single slice scanning

Multi-slice Spiral CT

- The efficiency is increased by incorporating an array of detectors in the head-foot direction
  - Modern multi-detector CT (MDCT) systems have up to 320 detector rows
- Higher values of the table feed per rotation
  - To image larger volumes in a given time, or
  - To image a given volume in a shorter scan time
- The slice thickness is dictated by the dimensions of the individual detectors, not the collimator

Dual-source and Dual-energy CT

- The temporal resolution of helical CT
  - Limited by the rotation rate of the gantry (~160 ms for a 180° rotation for existing systems)
  - Higher speed would generate higher forces on the system
  - A second set of X-ray tube/detectors (usually of a smaller FOV) can increase the temporal resolution for applications like cardiac imaging: dual source computed tomography (DSCT)



Dual-source and Dual-energy CT cont.

- 'Dual-energy' images
  - Produced by using a different kVp and tube current for each tube
  - To differentiate voxels which contain bone from those with contrast agent
  - Typically, the two kVp values used are 140 keV and 80 keV
  - Each voxel is decomposed into a percentage contribution from iodine, bone and soft-tissue
- Example: micro-calcifications on left image has been removed on right



Specialized Reconstruction Techniques

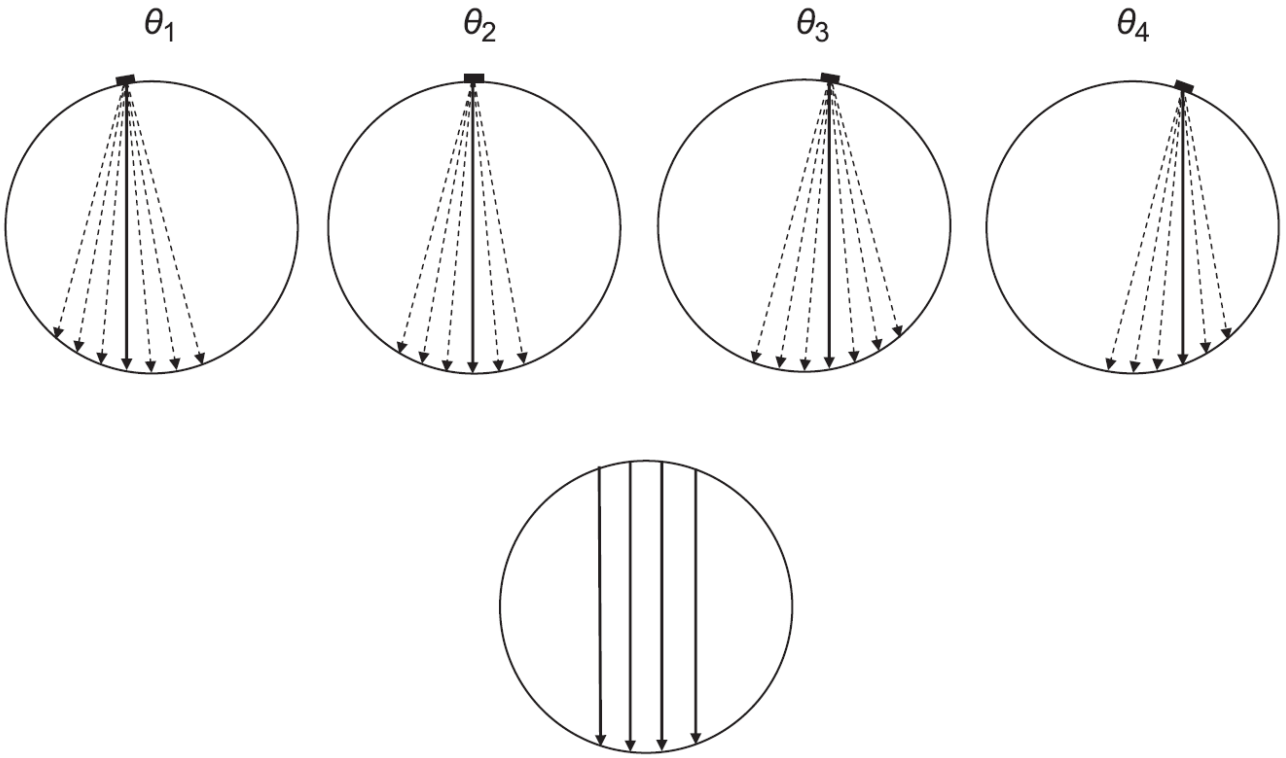
Fan Beam Reconstructions

- X-ray beam in CT
  - Not always parallel to the detectors, but filtered back-projection assumes a parallel beam
  - Subtends an angle of 45°-60° since the detector array is quite wide

- Synthetic projections
  - Selected set of projections from different angles act as parallel projections
  - Introduces negligible error with very high accuracy in knowing the position of at each time

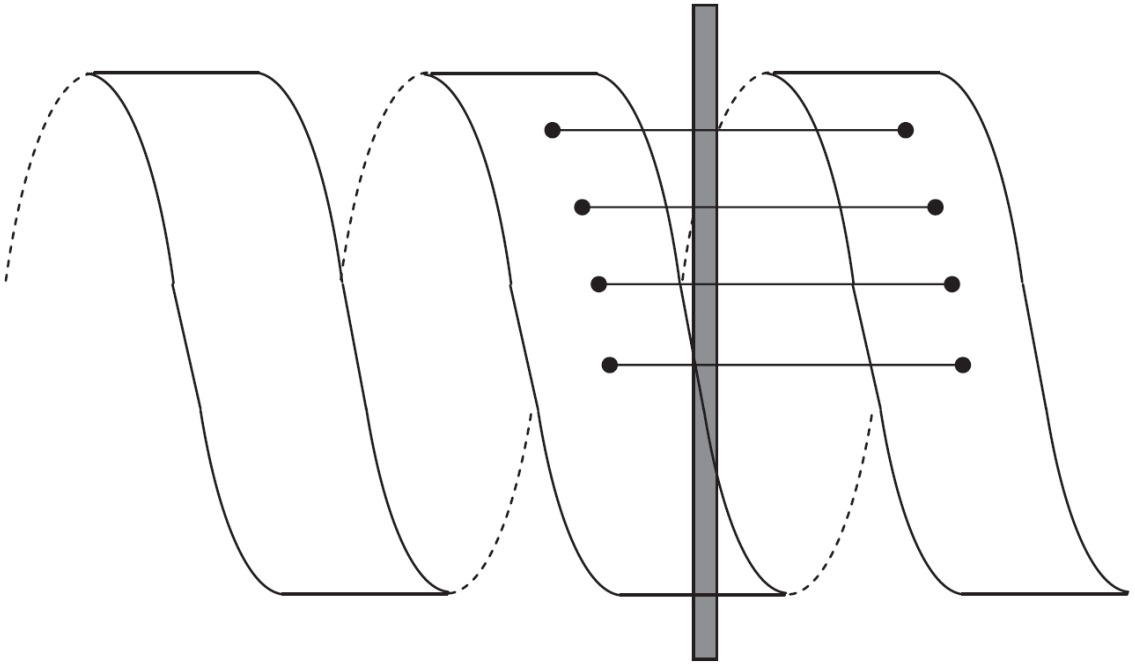
Fan Beam Reconstructions cont.

- Data re-binning to produce parallel projections from fan beam data
  - One projection (solid line) is taken from each dataset to produce a synthetic dataset (bottom) with four parallel projections



Reconstruction of Helical CT Data

- The helical trajectory of the X-rays through the patient
  - Does not correspond to a complete 180° or 360° rotation
  - Data must be resampled before filtered back-projection
  - The process involves interpolation of data points which are not acquired.



Reconstruction of Multi-slice Helical CT Scans

- Identical to single-slice helical scans

- Slice thickness can be chosen retrospectively after data acquisition using the same data
  - Thin slices can be reconstructed to form a **high resolution 3D image**
  - Thicker slices can be reconstructed to get a **higher SNR**

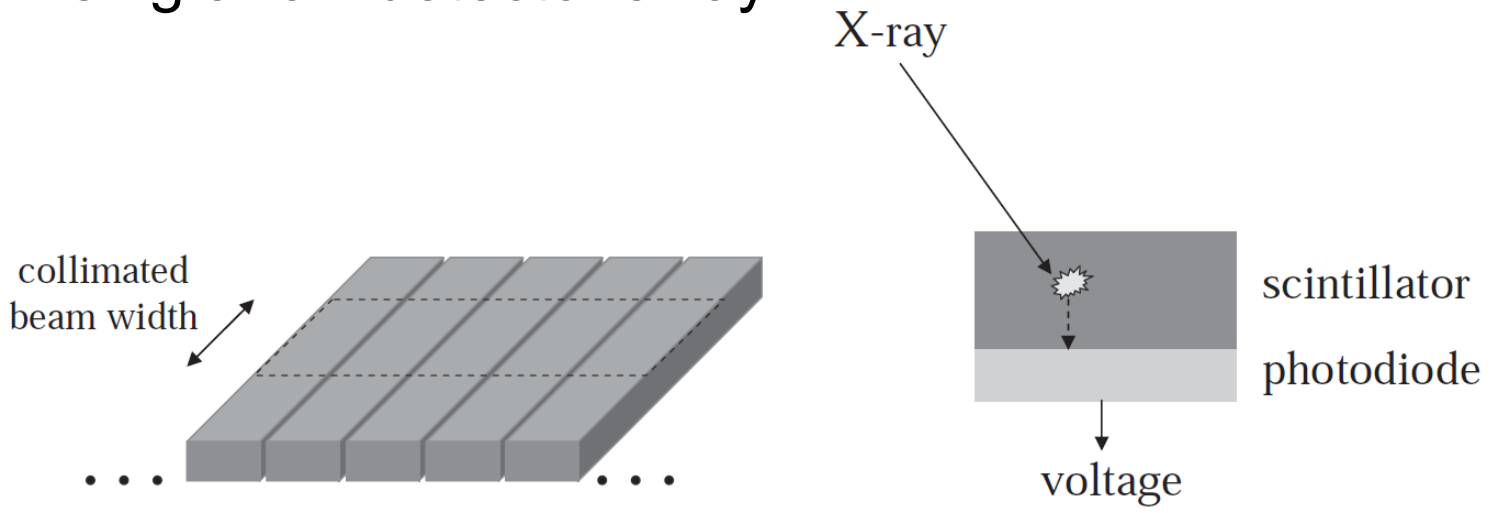
Instrumentation for CT

Instrumentation for CT

- Components of the CT system:
  - X-ray tube
  - Collimator
  - Anti-scatter grid
  - Detectors
- The components have to be fixed to a heavy gantry, which rotates very rapidly
  - High power cables are used to deliver power to the X-ray tube
  - A large parallel cable is used to transfer data from the scanner to the central computer
- Dedicated digital signal processing boards are used to pre-process the data from each detector
  - While the remainder of the data are being acquired
  - Speed up image reconstruction

Detectors used in CT

- Solid-state devices
  - Typically ~15 mm in length and 1 mm width
  - First convert the X-ray energy into light using a scintillator
  - Then light is converted into a voltage using a photo-diode
- The anti-scatter grid is usually integrated into the detector array
- Example: A single row detector array



Detectors for Multi-slice CT

- Multiple detector rows in the head/foot direction
  - Up to 320 separate rows of detectors, spanning a length of 16 cm
  - Separate signals are acquired from each of these detectors
  - The same number of circumferential elements per row as for single-detector helical CT
  - A large number of (~ 250, 000) separate signals being recorded essentially simultaneously
- Slice thickness is now determined by the size of the detector rather than the collimated beamwidth



Detectors for Multi-slice CT cont.

- Rows of the detectors
  - The dimensions of the separate rows can be the same or smaller detectors can be at the centre rows
- Example: In a 40 row detector array, the 32 centre rows having 0.6 mm width, and the eight outer rows having 1.2 mm width
  - Total coverage: 28.8 mm
  - Acquisition: 32 slices of 0.6 mm thickness at the centre of the array, or 24 slices of 1.2 mm thickness
- The z-flying spot technique
  - To improve the spatial resolution to a value smaller than the dimensions of the detector
  - The focal spot on the anode in the X-ray tube alternates rapidly between two slightly different positions

Inside the CT





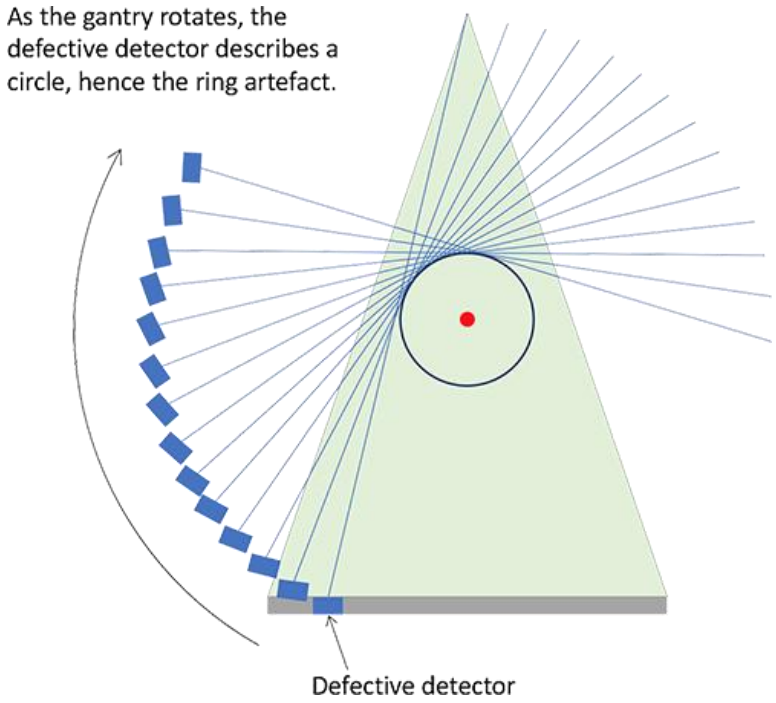
CT Artefacts

CT Artefacts

- An artefact is any discrepancy between the reconstructed and true attenuation coefficients
- Random noise due to photon statistics is not an artefact
- Artefacts are usually due to errors in measurement of X-ray attenuation
- Typically due to one of four causes
  - Faulty detectors(Miscalibration/Erroneous)
  - Scattered radiation
  - Beam Hardening (Spectral effects)
  - Faulty X-ray source (Very unlikely)

Faulty Detector (3rd Generation)

- Error in a detector causes a bad transmission measurement at each projection view
- Artefact is a ring of erroneous μ
- Error of only 0.1% can cause observable rings



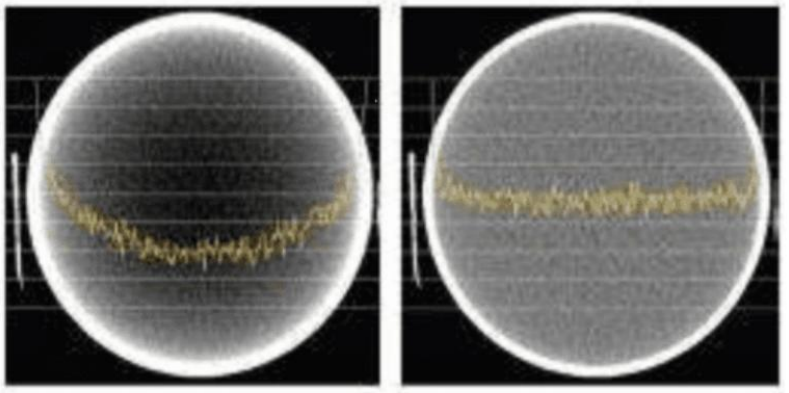


Faulty Detector (4th Generation)

- All rays in a single projection are bad
- This is a streak running across the image (Not a ring)

Scatter and Beam Hardening

- Some X-rays that should have been attenuated are scattered into other detectors
- Beam hardening results from selective attenuation of lower energy X-rays in X-ray beam, causing mean X-ray energy to increase
- Both scatter and beam hardening result in artificially large values of X-ray transmission
  - Overestimate in transmission results in underestimate of μ
  - Cupping effect



Left: Image with cupping artifact. Right: Image without cupping.

Clinical Applications

Clinical Applications

- Computed tomography is used for a wide range of clinical conditions in almost every organ of the body
  - Brain, lymph nodes, lung, pelvis, liver, gall bladder, spleen, kidneys, GI tract, spine, and extremities are all studied using CT

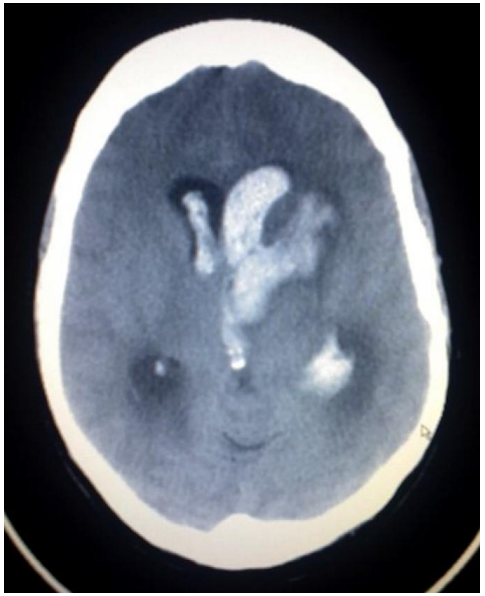
Cerebral Scans

- In cases of acute trauma
  - Signs of internal bleeding, tissue oedema (water accumulation) and skull fracture
  - Aneurysm ruptures leading to internal bleeding
  - Blood shows up as a high signal (high CT number) compared to the surrounding tissue, whereas oedema has a somewhat lower intensity than tissue

Normal



Haemorrage



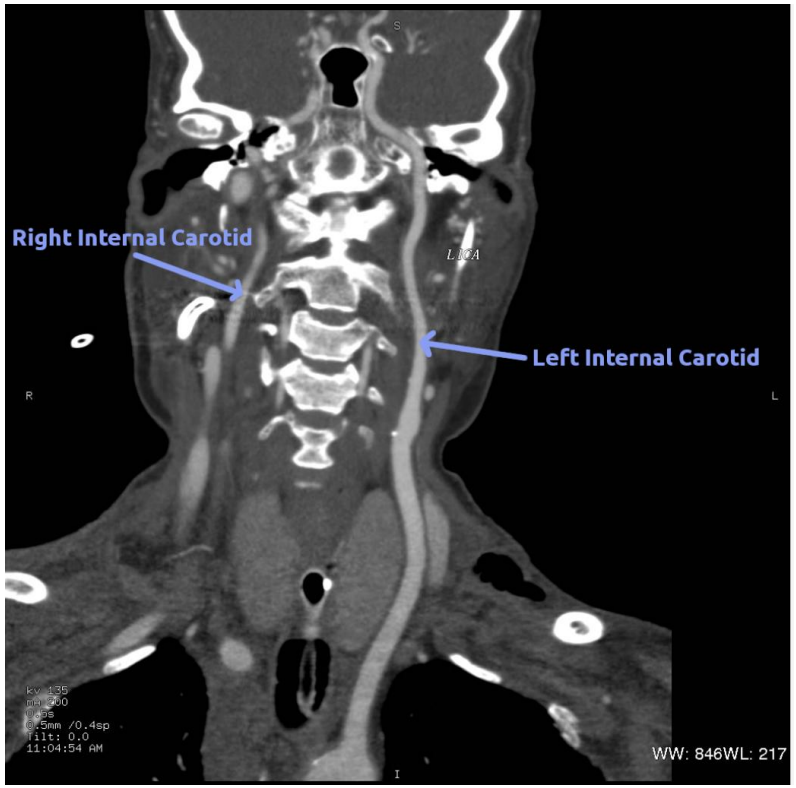
Cerebral Scans cont.

- To diagnose and to follow the progression of tumours
  - Meningiomas and gliomas are easy to diagnose

- In many cases, contrast agent is injected intravenously to highlight the tumour
  - In normal tissue: contrast agents do not cross the blood brain barrier (BBB)
  - Tumour growth: disrupts this barrier and so the agents can now enter the tumour

CT Angiography of the Head

- Uses an injection of iodine-based X-ray dye into a vein to see the vessels accurately
- Assesses the severity of any narrowing of the arteries



Chest CT

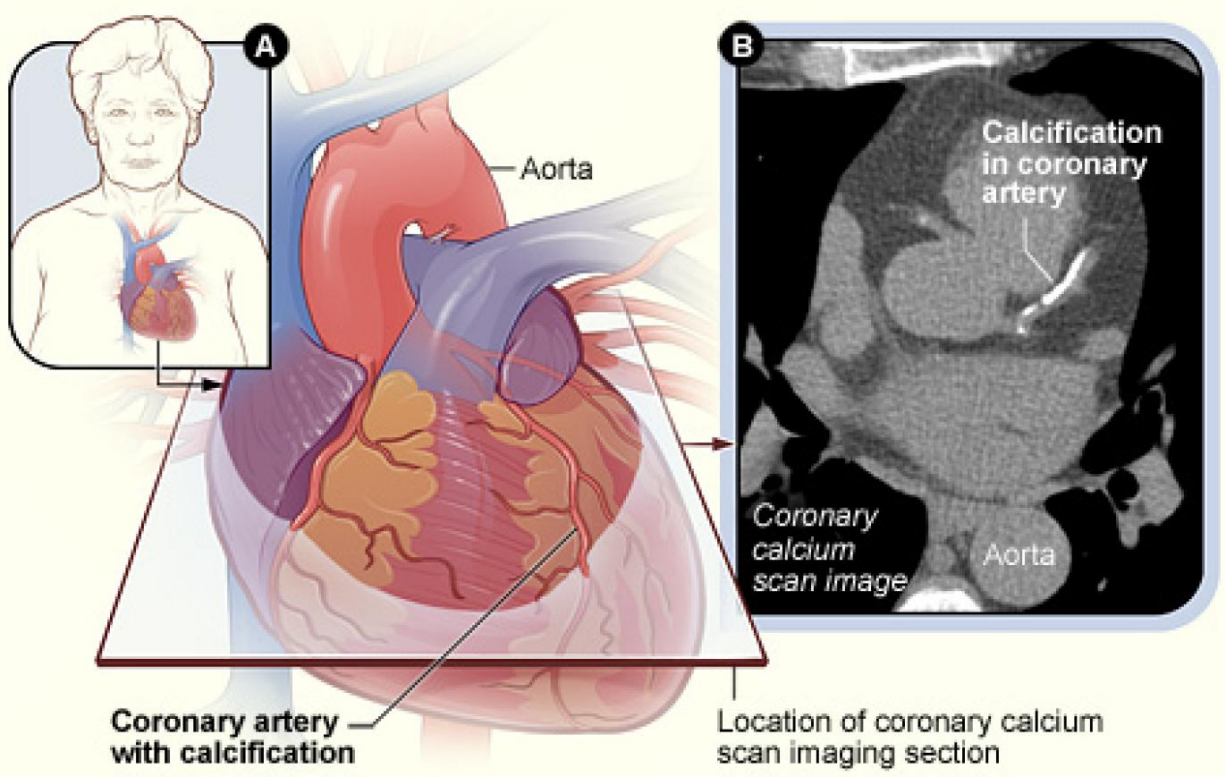
- To assess the chest and its organs
  - Respiratory and cardiovascular systems
  - To look for heart disease, aneurysms of the aorta, coronary artery calcification or pulmonary emboli
  - Oesophagus, for tumours and other lesions, intra-thoracic bleeding, infections, obstructions, or other conditions

Cardiac Imaging

- Fast image acquisition to ‘freeze’ the cardiac motion
  - Acquire all the projections within a single breath-hold if possible
  - Image acquisition is gated to the heart motion via an ECG trigger

Cardiac Imaging cont.

- Coronary Artery Calcification Scoring (CACS) CT scan
  - A non-contrast scan
  - Calcification within the coronary artery wall can be an indicator of the presence of coronary artery disease

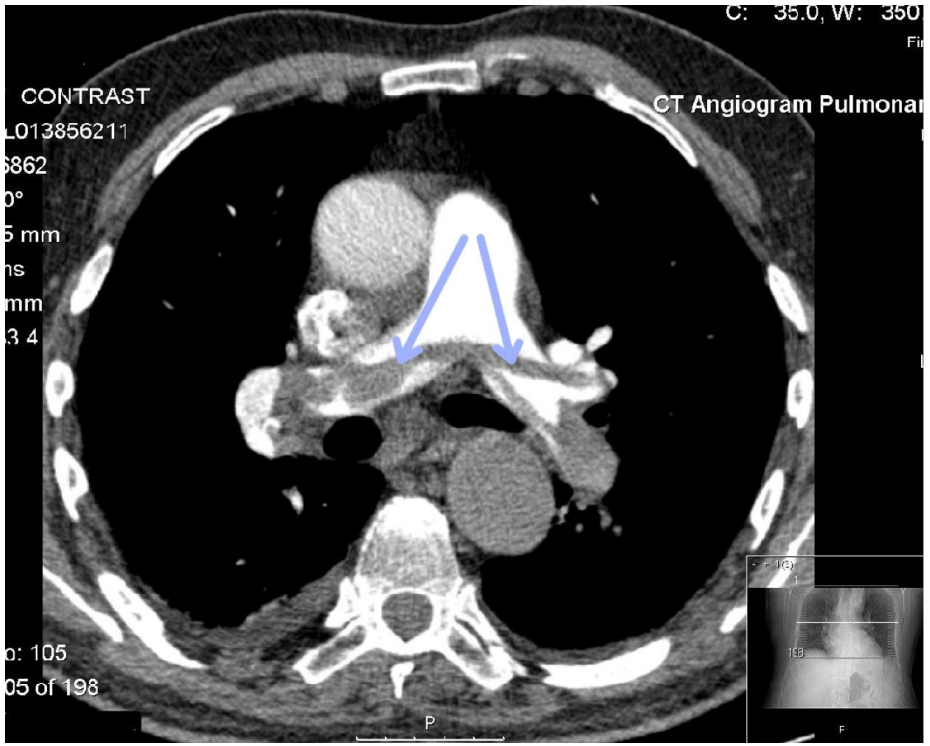


Pulmonary CT

- Normal lung tissue appears dark on CT images
  - But not as dark as the air
  - A CT scan can outline the size and geometry of a lesion
  - Lung cancer produces nodules with irregular edge
  - Calcification of a nodule normally indicates a benign rather than metastatic tumour
  - Emphysema shows up as large areas of low attenuation compared to the surrounding

Pulmonary CT cont.

- During a CT pulmonary angiogram
  - An intravenous injection of an iodine-containing contrast agent is used
  - Images are acquired with the maximum intensity of radio-opaque contrast in the pulmonary arteries
  - Example: the grey matter indicated by the arrows is a blood clot



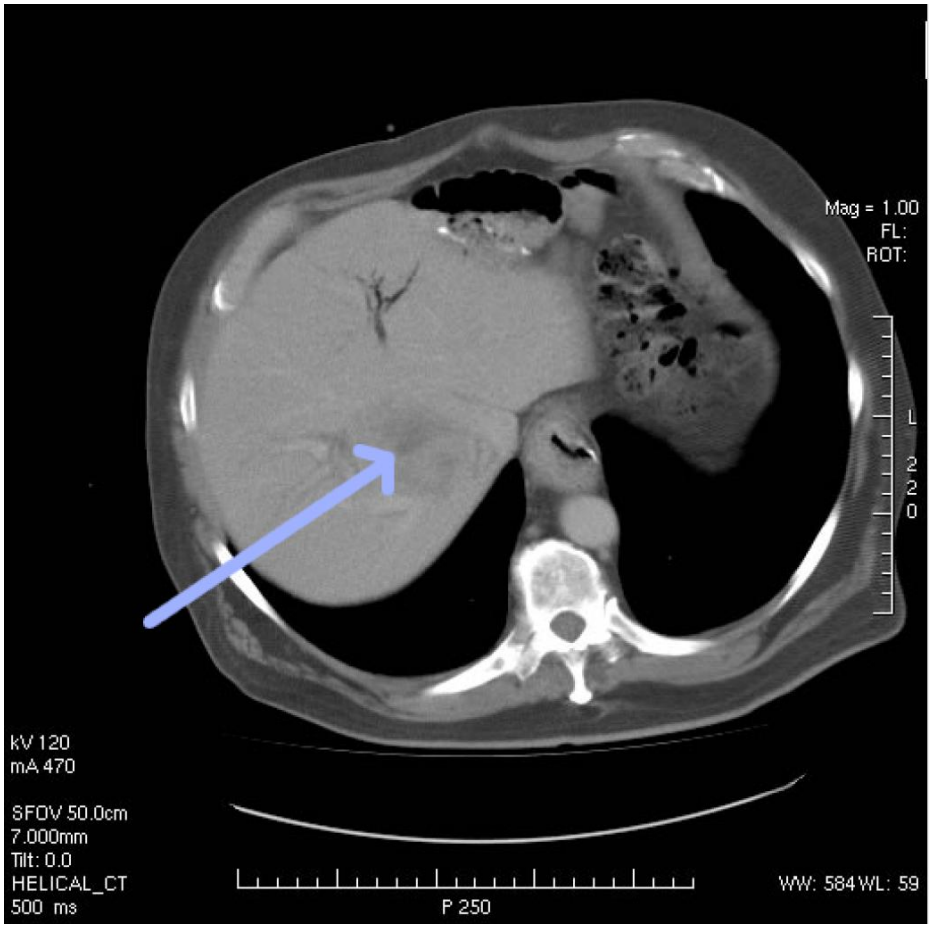
Abdomen CT

- Many organs inside the abdomen which may be affected by cancer, infection, perforation and trauma due to injury

- Organs which may be analysed using CT include liver, large bowel, small bowel, kidneys, stomach, pancreas, gallbladder, uterus, bladder, ovaries and spleen

Abdomen CT cont.

- Example: CT image of an abdominal scan (axial view) showing liver abscess (blue arrow)



Abdomen CT cont.

- Example: CT image of an abdominal scan (axial view) showing kidney stone (blue arrow)



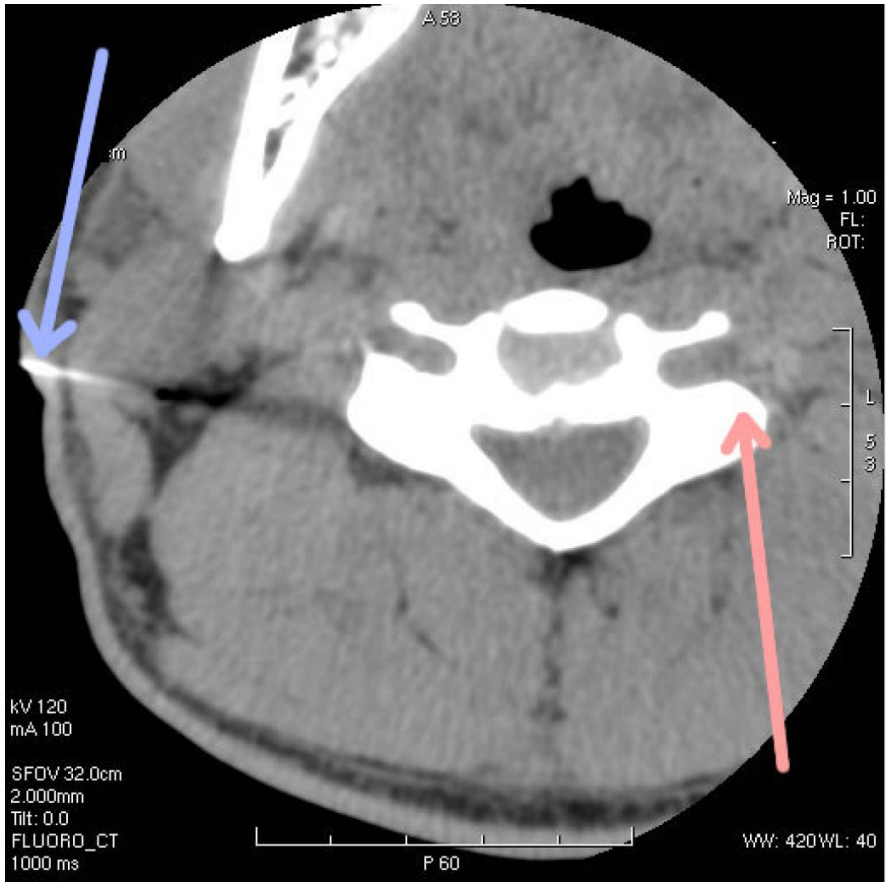
Spine CT

- Cross sectional images of the spine and spinal cord
  - Provides information about the vertebrae (bones of the spine) and other spinal structures and tissues

- Performed to assess the spine for:
  - Disk hernia
  - Tumours and other lesions
  - Spinal cord compression
  - Structural anomalies, such as spina bifida (a congenital defect)

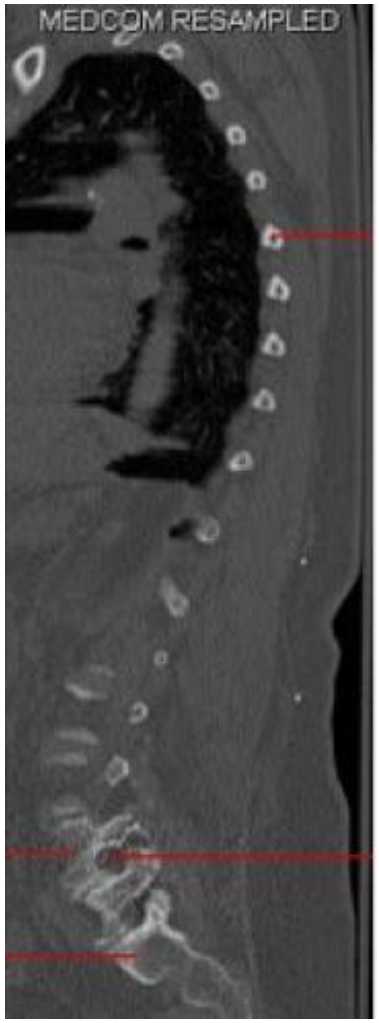
Spine CT cont.

- Example: CT image of an upper cervical spine scan (axial view) showing a needle (blue arrow) delivering an anesthetic injection in a facet joint of the vertebrae (red arrow).

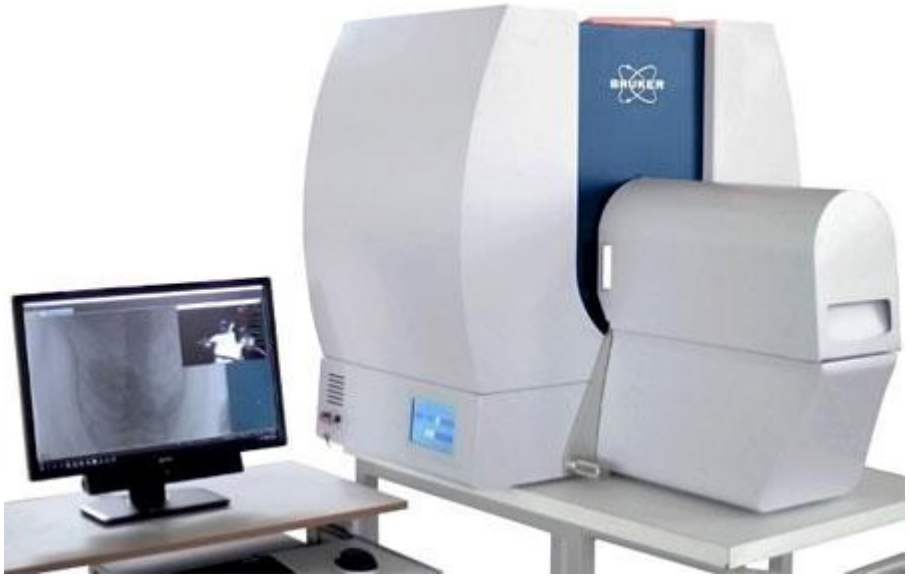


Spine CT cont.

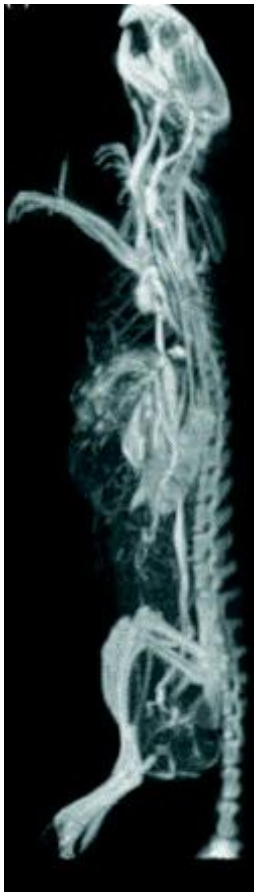
- Example: CT image of a Lumbo-sacral spine scan (sagittal view)

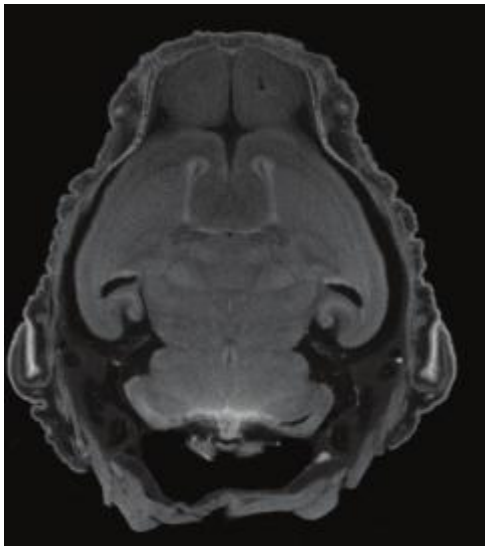


CT of Animals



Micro-CT Scanners





Spatial resolution
~ 5 μm

CT Dose Considerations

- Dose is typically higher than radiography
- Normal yearly background dose is **2-3** mSv
- Dose for a typical chest X-ray is less than **0.1** mSv

Anatomic Region	**Organ Dose (mSv)**	**Effective Dose (mSv)**
Head	28.1	1.1
Chest	23.3	6.7
Abdomen	12.9	4.3
Pelvis	13.3	2.7